Lecture 3, Appendix: Topology, Differential Topology and Functional Analysis

Math for Machine Learning

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Introduction

In Lecture 3, we used several facts without proof: that a continuous function on a closed and bounded set attains its maximum, that a function with continuous partial derivatives is differentiable (Theorem 3.6), that integrals depending on a parameter are continuous (in the proof of Proposition 14.6), the implicit function theorem in the proof of the Lagrange multiplier theorem (Theorem 12.1), the necessity of the KKT conditions (Theorem 12.5), and the existence and uniqueness of solutions of differential equations (Theorem 15.2). We also met sets that are not open subsets of $\mathbb{R}^n$, but curved objects: spheres, the set of orthogonal matrices, and level sets of functions. And in Lectures 2 and 4, we worked with vector spaces of functions and of random variables, which are infinite-dimensional. This appendix provides the mathematical background.

- Sections A.1–A.4 are about topology, the mathematics of nearness and continuity. Its central notions are open sets, compactness and connectedness. Topology studies the properties of sets that do not change under continuous deformations.
- Sections A.5–A.10 are about differential topology, which studies manifolds, i.e., sets that look like $\mathbb{R}^k$ near each of their points, and differentiable maps between them. Its main tools are the inverse and implicit function theorems. Morse theory connects the critical points of a function, classified by the eigenvalues of its Hessian as in Lecture 3, with the shape of the space on which it is defined. Finally, differential forms, which are built from the tensors of Lecture 1, unify the integral theorems of Lecture 3 in one theorem of Stokes, and they detect holes in a space.
- Section A.11 collects some places where these ideas appear in machine learning.
- Sections A.12–A.15 are about functional analysis, i.e., linear algebra and analysis in infinite-dimensional vector spaces, such as spaces of functions. Its central notions are completeness, orthogonal projections and the spectral theorem for compact operators. They prove the existence theorem for differential equations of Lecture 3 and the Radon–Nikodym theorem of the appendix to Lecture 4, and they lead to Fourier series, to expansions of random functions, and to the theory of kernel methods. Section A.16 returns to machine learning.

We use the notation of Lecture 3: vectors in $\mathbb{R}^n$ are bold, and $\|\cdot\|$ is the Euclidean norm unless we say otherwise. Elements of general vector spaces, such as functions, are written in normal font.

A.1 Metric Spaces and Open Sets

Definition A.1.1 (Metric space)**.** A metric on a set X is a function $d : X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:

1. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$;
2. $d(x, y) = d(y, x)$;
3. $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

A set with a metric is a metric space.

Example A.1.2.
1. Every norm on a vector space defines a metric $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}-\mathbf{y}\|$ (Lecture 2). In particular, $\mathbb{R}^n$ with the ℓ_1, ℓ_2 or ℓ_∞ norm is a metric space.
2. On the space $C[a, b]$ of continuous functions on $[a, b]$, $d(f, g) = \max_{x\in[a,b]} |f(x) - g(x)|$ is a metric.
3. Every subset of a metric space is a metric space with the same metric. For example, the unit sphere $S^{n-1} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = 1\}$ is a metric space with the Euclidean distance.
4. On any set, the discrete metric $d(x, y) = 1$ for $x \neq y$ (and $d(x, x) = 0$) is a metric.

Definition A.1.3 (Open and closed sets)**.** Let (X, d) be a metric space. The open ball with center x and radius $r > 0$ is $B(x, r) = \{y \in X : d(x, y) < r\}$. A set $U \subseteq X$ is open if for every $x \in U$ there is an $r > 0$ with $B(x, r) \subseteq U$. A set $C \subseteq X$ is closed if its complement $X \setminus C$ is open.

For $X = \mathbb{R}^n$, this is the definition of open sets in Lecture 3. Closed is not the opposite of open: $\mathbb{R}^n$ and $\emptyset$ are both open and closed, and the half-open interval $[0, 1)$ is neither open nor closed in $\mathbb{R}$.

Proposition A.1.4. *In a metric space X:*

1. *every open ball is open;*
2. *every union of open sets is open, and every intersection of finitely many open sets is open;*
3. *every intersection of closed sets is closed, and every union of finitely many closed sets is closed.*

Proof. (1) If $y \in B(x, r)$, let $s = r - d(x, y) > 0$. For $z \in B(y, s)$, the triangle inequality gives $d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + s = r$, so $B(y, s) \subseteq B(x, r)$.

(2) If x lies in a union of open sets, it lies in one of them, U, and some ball $B(x, r) \subseteq U$ lies in the union. If $x \in U_1 \cap \cdots \cap U_m$ with $B(x, r_i) \subseteq U_i$, then $B(x, \min_i r_i)$ lies in the intersection.

(3) Take complements in (2). □

Infinite intersections of open sets need not be open: $\bigcap_{k\geq1}(-\frac{1}{k}, \frac{1}{k}) = \{0\}$. For a set $A \subseteq X$, the interior $\operatorname{int} A$ is the union of all open sets contained in A (the largest open subset of A), the closure $\overline{A}$ is the intersection of all closed sets containing A (the smallest closed set containing A), and the boundary is $\partial A = \overline{A} \setminus \operatorname{int} A$. For example, the closed ball $\{\mathbf{x} : \|\mathbf{x}\| \leq 1\}$ in $\mathbb{R}^n$ has the open ball as interior and the sphere S^{n-1} as boundary.

Definition A.1.5 (Convergence)**.** A sequence (x_k) in a metric space converges to x if $d(x_k, x) \to 0$ as $k \to \infty$. It is a Cauchy sequence if for every $\varepsilon > 0$ there is an N with $d(x_k, x_l) < \varepsilon$ for all $k, l \geq N$.

Closed sets are exactly the sets that cannot be left by taking limits.

Proposition A.1.6. *A subset C of a metric space X is closed if and only if the limit of every convergent sequence of points of C lies in C.*

Proof. Let C be closed, and let $x_k \in C$ converge to x. If $x \notin C$, then some ball $B(x, r)$ lies in the open set $X \setminus C$; but $x_k \in B(x, r)$ for large k, a contradiction. Conversely, suppose that C is not closed. Then $X \setminus C$ is not open, so there is an $x \notin C$ such that every ball $B(x, \frac{1}{k})$ contains a point $x_k \in C$. Then $x_k \to x$, and $x \notin C$. □

A.2 Continuity and Topological Spaces

Continuity can be described with open sets alone, without mentioning distances.

Proposition A.2.1. *Let $f : X \to Y$ be a map between metric spaces. Then f is continuous (at every point, in the ε-δ sense of Lecture 3) if and only if for every open set $V \subseteq Y$, the preimage $f^{-1}(V) = \{x \in X : f(x) \in V\}$ is open.*

Proof. Let f be continuous, $V \subseteq Y$ open, and $x \in f^{-1}(V)$. There is an $\varepsilon > 0$ with $B(f(x), \varepsilon) \subseteq V$, and by continuity a $\delta > 0$ such that $d(x, x') < \delta$ implies $d(f(x), f(x')) < \varepsilon$, i.e., $B(x, \delta) \subseteq f^{-1}(V)$. Conversely, if preimages of open sets are open, then for $x \in X$ and $\varepsilon > 0$, the set $f^{-1}(B(f(x), \varepsilon))$ is open and contains x, so it contains a ball $B(x, \delta)$; this is the ε-δ condition. □

Taking complements, f is continuous if and only if preimages of closed sets are closed.

Corollary A.2.2. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous and $c \in \mathbb{R}$. Then the set $\{f < c\}$ is open, and the sublevel set $\{f \leq c\}$ and the level set $\{f = c\}$ are closed. The same holds for intersections of finitely many such sets.*

Proof. These are the preimages of the open set $(-\infty, c)$ and the closed sets $(-\infty, c]$ and $\{c\}$. □

So the feasible sets of the constrained problems in Lecture 3, such as $\{\mathbf{x} : g_1(\mathbf{x}) \leq 0, \ldots, g_m(\mathbf{x}) \leq 0\}$ for continuous g_i, are closed. Similarly, the set $\mathrm{GL}(n, \mathbb{R})$ of invertible $n \times n$ matrices is open in $\mathbb{R}^{n \times n}$, as the preimage of the open set $\mathbb{R} \setminus \{0\}$ under the continuous function det (a polynomial in the entries).

Proposition A.2.1 suggests an abstraction: forget the metric, and only remember which sets are open.

Definition A.2.3 (Topological space)**.** A topology on a set X is a collection $\mathcal{T}$ of subsets of X, called open sets, such that $\emptyset, X \in \mathcal{T}$, every union of sets in $\mathcal{T}$ lies in $\mathcal{T}$, and every intersection of finitely many sets in $\mathcal{T}$ lies in $\mathcal{T}$. A map $f : X \to Y$ between topological spaces is continuous if preimages of open sets are open. A homeomorphism is a continuous bijection whose inverse is also continuous; if one exists, X and Y are homeomorphic.

By Proposition A.1.4, the open sets of a metric space form a topology. Two metrics can give the same topology: we will see in Theorem A.3.10 that all norms on $\mathbb{R}^n$ do. Homeomorphic spaces are the same from the point of view of topology: a homeomorphism matches up their open sets, convergent sequences, continuous functions, and so on.

Example A.2.4. 1. The sigmoid function is a homeomorphism from $\mathbb{R}$ to $(0,1)$, with the continuous inverse $p \mapsto \log \frac{p}{1-p}$ (the logit). Similarly, tanh is a homeomorphism $\mathbb{R} \to (-1,1)$. So bounded and unbounded sets can be homeomorphic: lengths and distances are not topological properties.

2. The open ball $B(\mathbf{0},1) \subseteq \mathbb{R}^n$ is homeomorphic to $\mathbb{R}^n$, via $\mathbf{x} \mapsto \frac{\mathbf{x}}{1-\|\mathbf{x}\|}$, with the inverse $\mathbf{y} \mapsto \frac{\mathbf{y}}{1+\|\mathbf{y}\|}$.

3. A square and a disk in $\mathbb{R}^2$ are homeomorphic (rescale each ray from the center), and so are a circle and the boundary of a square: corners are not a topological property. But a circle and an interval are not homeomorphic (Example A.4.3).

A.3 Compactness

The real numbers have no gaps. This is the completeness axiom, which we take as one of the defining properties of $\mathbb{R}$: every nonempty set of real numbers that is bounded above has a least upper bound (supremum). It implies that every bounded monotone sequence of real numbers converges (to the supremum or the infimum of its values).

Theorem A.3.1 (Bolzano–Weierstrass)**.** *Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.*

Proof. First let $n = 1$. Call an index k a peak if $x_k \geq x_l$ for all $l > k$. If there are infinitely many peaks $k_1 < k_2 < \cdots$, then $x_{k_1} \geq x_{k_2} \geq \cdots$ is a monotone subsequence. Otherwise, there is an index k_1 after which there are no peaks; then k_1 is not a peak, so $x_{k_2} > x_{k_1}$ for some $k_2 > k_1$; k_2 is not a peak either, and so on. This gives an increasing subsequence. In both cases, the monotone subsequence is bounded, so it converges. For $n > 1$, choose a subsequence whose first coordinates converge, then a subsequence of it whose second coordinates converge, and so on; after n steps all coordinates converge. □

Corollary A.3.2. *$\mathbb{R}^n$ is complete: every Cauchy sequence in $\mathbb{R}^n$ converges.*

Proof. A Cauchy sequence $(\mathbf{x}_k)$ is bounded (all terms after some N lie within distance 1 of $\mathbf{x}_N$), so by Theorem A.3.1 a subsequence converges to some $\mathbf{x}$. For $\varepsilon > 0$, choose N such that $\|\mathbf{x}_k - \mathbf{x}_l\| < \varepsilon$ for $k, l \geq N$, and an element $\mathbf{x}_l$ of the subsequence with $l \geq N$ and $\|\mathbf{x}_l - \mathbf{x}\| < \varepsilon$. Then $\|\mathbf{x}_k - \mathbf{x}\| < 2\varepsilon$ for all $k \geq N$. □

Definition A.3.3 (Compact set)**.** A subset K of a metric space is compact if every sequence in K has a subsequence that converges to a point of K.

(In general topological spaces, one defines compactness with open covers: K is compact if every collection of open sets that covers K contains finitely many sets that already cover K. For metric spaces, the two definitions agree; we do not prove this.)

Theorem A.3.4 (Heine–Borel)**.** *A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proof. Let K be closed and bounded, and let $(\mathbf{x}_k)$ be a sequence in K. By Theorem A.3.1, a subsequence converges, and by Proposition A.1.6, its limit lies in K. Conversely, if K is unbounded, there are $\mathbf{x}_k \in K$ with $\|\mathbf{x}_k\| \geq k$, and no subsequence converges, since convergent sequences are bounded. If K is not closed, there is a sequence in K that converges to a point outside K (Proposition A.1.6), and all its subsequences converge to the same point, so none converges to a point of K. □

For example, spheres, closed balls (for any norm), and the set $\mathrm{O}(n)$ of orthogonal matrices are compact. $\mathrm{O}(n)$ is closed, as the preimage of $\{I\}$ under the continuous map $Q \mapsto Q^\top Q$, and bounded, since $\|Q\|_F^2 = \operatorname{tr}(Q^\top Q) = n$. The hyperbola $\{xy = 1\}$ is closed but unbounded, and the open ball is bounded but not closed; neither is compact.

Theorem A.3.5 (Extreme value theorem)**.** *Let K be a nonempty compact set, and let $f : K \to \mathbb{R}$ be continuous. Then f attains a minimum and a maximum on K.*

Proof. Let $m = \inf\{f(\mathbf{x}) : \mathbf{x} \in K\}$, where we allow $m = -\infty$. Choose $\mathbf{x}_k \in K$ with $f(\mathbf{x}_k) \to m$. A subsequence converges to some $\mathbf{x}^* \in K$, and by continuity, $f(\mathbf{x}^*) = m$. So m is finite, and it is the minimum. For the maximum, apply this to $-f$. □

This justifies the claims in Lecture 3 that $f(x, y) = x + y$ and the Rayleigh quotient $\mathbf{x}^\top A\mathbf{x}$ attain their maxima on the unit sphere (Examples 12.2 and 12.3 of Lecture 3). On $\mathbb{R}^n$, which is not compact, a minimum need not exist, but the following condition is often enough.

Proposition A.3.6 (Coercive functions)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous and coercive, i.e., $f(\mathbf{x}) \to \infty$ as $\|\mathbf{x}\| \to \infty$. Then f has a global minimum.*

Proof. Fix any $\mathbf{x}_0$. The sublevel set $S = \{f \le f(\mathbf{x}_0)\}$ is closed (Corollary A.2.2) and bounded (by coercivity), so it is compact and nonempty, and f has a minimum $\mathbf{x}^*$ on S. Points outside S have values larger than $f(\mathbf{x}_0) \ge f(\mathbf{x}^*)$, so $\mathbf{x}^*$ is a global minimum. □

Example A.3.7. A loss $L(\mathbf{w}) \ge 0$ plus a ridge penalty, $L(\mathbf{w}) + \lambda\|\mathbf{w}\|^2$ with $\lambda > 0$, is coercive, so it has a global minimum whenever L is continuous. Without regularization, a minimum need not exist: for logistic regression with the single training example $x = 1$, $y = 1$, the loss $\log(1 + e^{-w})$ decreases towards 0 as $w \to \infty$, but never reaches it. The same happens for every linearly separable data set, where scaling up a separating weight vector always decreases the loss.

On a compact set, continuity is automatically uniform.

Proposition A.3.8 (Uniform continuity)**.** *Let $K \subseteq \mathbb{R}^n$ be compact, and let $f : K \to \mathbb{R}$ be continuous. Then f is uniformly continuous: for every $\varepsilon > 0$, there is a $\delta > 0$ such that $|f(\mathbf{x}) - f(\mathbf{y})| < \varepsilon$ for all $\mathbf{x}, \mathbf{y} \in K$ with $\|\mathbf{x} - \mathbf{y}\| < \delta$.*

Proof. Otherwise, there are an $\varepsilon > 0$ and points $\mathbf{x}_k, \mathbf{y}_k \in K$ with $\|\mathbf{x}_k - \mathbf{y}_k\| < \frac{1}{k}$ and $|f(\mathbf{x}_k) - f(\mathbf{y}_k)| \ge \varepsilon$ for every k. A subsequence $(\mathbf{x}_{k_l})$ converges to some $\mathbf{x} \in K$, and then $\mathbf{y}_{k_l} \to \mathbf{x}$ as well. By continuity, $f(\mathbf{x}_{k_l}) - f(\mathbf{y}_{k_l}) \to f(\mathbf{x}) - f(\mathbf{x}) = 0$, which contradicts $|f(\mathbf{x}_{k_l}) - f(\mathbf{y}_{k_l})| \ge \varepsilon$. □

The point is that δ depends only on ε, not on the points. On a set that is not compact, this can fail: $f(x) = x^2$ is continuous on $\mathbb{R}$, but $f(x + \delta) - f(x) = 2x\delta + \delta^2$ is large for large x, however small $\delta > 0$ is.

Corollary A.3.9 (Integrals depending on a parameter)**.** *Let $h(x, \theta)$ be continuous for $x \in [a, b]$ and θ in an open interval I. Then $H(\theta) = \int_a^b h(x, \theta)\, dx$ is continuous on I.*

Proof. Fix $\theta_0 \in I$, and choose $r > 0$ with $[\theta_0 - r, \theta_0 + r] \subseteq I$. The set $K = [a, b] \times [\theta_0 - r, \theta_0 + r]$ is closed and bounded, hence compact, so h is uniformly continuous on K. For $\varepsilon > 0$, choose $\delta \le r$ as in Proposition A.3.8. If $|\theta - \theta_0| < \delta$, then (x, θ) and (x, θ_0) have the distance $|\theta - \theta_0| < \delta$ for every x, so

$$|H(\theta) - H(\theta_0)| \le \int_a^b |h(x, \theta) - h(x, \theta_0)|\, dx \le (b - a)\varepsilon.$$ □

Applied to $h = \frac{\partial f}{\partial \theta}$, this fills the gap in the proof of the Leibniz rule (Proposition 14.6 of Lecture 3).

Compactness also explains why all norms on $\mathbb{R}^n$ lead to the same notions of convergence and continuity, as claimed in Lecture 3.

Theorem A.3.10 (Equivalence of norms)**.** *For every norm N on $\mathbb{R}^n$, there are constants $0 < c \le C$ with $c\|\mathbf{x}\| \le N(\mathbf{x}) \le C\|\mathbf{x}\|$ for all $\mathbf{x}$. Consequently, all norms on $\mathbb{R}^n$ define the same open sets.*

Proof. Writing $\mathbf{x} = \sum_i x_i \mathbf{e}_i$, the triangle inequality and the Cauchy–Schwarz inequality give

$$N(\mathbf{x}) \le \sum_i |x_i| N(\mathbf{e}_i) \le \Big(\sum_i N(\mathbf{e}_i)^2 \Big)^{1/2} \|\mathbf{x}\| = C\|\mathbf{x}\|.$$

In particular, $|N(\mathbf{x}) - N(\mathbf{y})| \le N(\mathbf{x} - \mathbf{y}) \le C\|\mathbf{x} - \mathbf{y}\|$, so N is continuous. By Theorem A.3.5, N attains a minimum c on the compact unit sphere S^{n-1}, and $c > 0$ since N is positive on S^{n-1}. For $\mathbf{x} \neq \mathbf{0}$, $N(\mathbf{x}) = \|\mathbf{x}\| \, N\big(\frac{\mathbf{x}}{\|\mathbf{x}\|}\big) \ge c\|\mathbf{x}\|$. Finally, the inequalities show that every N-ball around a point contains a Euclidean ball around it and vice versa, so the open sets are the same. □

Remark A.3.11 (Infinite dimensions)*.* In infinite-dimensional spaces, closed and bounded sets need not be compact, and different norms can give different topologies. For example, in the space ℓ^2 of square-summable sequences (Section A.12), the unit sequences $e_1, e_2, \ldots$ lie in the closed unit ball, but $\|e_j - e_k\| = \sqrt{2}$ for $j \neq k$, so no subsequence converges. This is one reason why analysis in function spaces, such as the spaces of functions in Lecture 2, is harder than in $\mathbb{R}^n$ (Sections A.12–A.15).

A.4 Connectedness

Definition A.4.1 (Connected sets)**.** A subset X of a metric space is disconnected if $X \subseteq U \cup V$ for two open sets U and V such that $X \cap U$ and $X \cap V$ are nonempty and disjoint. Otherwise, X is connected. It is path-connected if any two points $x, y \in X$ can be joined by a path, i.e., a continuous map $\gamma : [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$.

Proposition A.4.2. *1. Every interval of $\mathbb{R}$ is connected.*

2. If X is connected and f is continuous on X, then $f(X)$ is connected.

3. Every path-connected set is connected.

Proof. (1) Suppose an interval I is covered by open sets U and V as in the definition, with $a \in I \cap U$ and $b \in I \cap V$; by symmetry, we may assume $a < b$. Then $[a, b] \subseteq I$. Let $s = \sup\{t \in [a, b] : [a, t] \subseteq U\}$; then $[a, t] \subseteq U$ for every $t < s$, and $s \in [a, b] \subseteq I$, so s lies in U or in V. If $s \in V$, then since V is open, $(s - \varepsilon, s] \subseteq V$ for some $\varepsilon > 0$; but points of $(s - \varepsilon, s)$ also lie in U, which contradicts $I \cap U \cap V = \emptyset$. If $s \in U$, then $s \neq b$ (since $b \in V$), so $s < b$, and since U is open, $[s, s + \varepsilon] \subseteq U$ for some small $\varepsilon > 0$ with $s + \varepsilon \le b$. Then $[a, s + \varepsilon] \subseteq U$, which contradicts the definition of s.

(2) If $f(X)$ were covered by open sets U and V as in the definition, then the preimages $f^{-1}(U)$ and $f^{-1}(V)$ would disconnect X (they are open by continuity, in the sense that they are of the form $X \cap U'$ for open sets U').

(3) Suppose that open sets U, V disconnect a path-connected set X, with $x \in U$ and $y \in V$. Then the preimages of U and V under a path from x to y would disconnect the interval $[0, 1]$, which contradicts (1). □

Together, (1) and (2) give the intermediate value theorem: a continuous function $f : [a, b] \to \mathbb{R}$ takes every value between $f(a)$ and $f(b)$, since its image is a connected subset of $\mathbb{R}$, i.e., an interval.

Example A.4.3.

1. $\mathbb{R}^n$, balls, and the sphere S^{n-1} for $n \geq 2$ are path-connected, hence connected.
2. The hyperbola $\{xy = 1\}$ is not connected: it lies in the union of the open half-planes $\{x > 0\}$ and $\{x < 0\}$, and meets both.
3. $\mathrm{GL}(n, \mathbb{R})$ is not connected: it is the union of the disjoint open sets $\{\det > 0\}$ and $\{\det < 0\}$. In particular, an invertible matrix with $\det < 0$ (a reflection) cannot be continuously deformed into the identity through invertible matrices. Similarly, $\mathrm{O}(n)$ consists of the two pieces $\det Q = 1$ (the rotations, Lecture 2) and $\det Q = -1$; one can show that the rotations form a connected set.
4. A circle and an interval are not homeomorphic. If we remove a point from a circle, the rest is still connected (it is an arc); if we remove an interior point from an interval, the rest is disconnected. A homeomorphism would map a point and its complement to a point and its complement, and it preserves connectedness by Proposition A.4.2(2).

A.5 The Inverse Function Theorem

From now on, we combine topology with derivatives. A map is smooth if all its partial derivatives of all orders exist and are continuous, and a diffeomorphism is a smooth bijection with a smooth inverse. We first fill a gap from Lecture 3, using a vector-valued version of the mean value theorem.

Lemma A.5.1 (Mean value inequality)**.** *Let $\mathbf{g} : U \to \mathbb{R}^m$ be continuously differentiable on an open set $U \subseteq \mathbb{R}^n$ that contains the segment between $\mathbf{x}$ and $\mathbf{y}$. Then*

$$\|\mathbf{g}(\mathbf{y}) - \mathbf{g}(\mathbf{x})\| \leq \sup_{\mathbf{z} \in [\mathbf{x}, \mathbf{y}]} \|D\mathbf{g}(\mathbf{z})\|_2 \, \|\mathbf{y} - \mathbf{x}\|,$$

where $\|\cdot\|_2$ is the spectral norm of Lecture 2.

Proof. Let $\mathbf{u} = \mathbf{g}(\mathbf{y}) - \mathbf{g}(\mathbf{x})$, and assume $\mathbf{u} \neq \mathbf{0}$. The function $\psi(t) = \mathbf{u}^\top \mathbf{g}(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$ has $\psi(1) - \psi(0) = \|\mathbf{u}\|^2$. By the mean value theorem and the chain rule, $\psi(1) - \psi(0) = \psi'(\tau) = \mathbf{u}^\top D\mathbf{g}(\mathbf{z})(\mathbf{y} - \mathbf{x})$ with $\mathbf{z} = \mathbf{x} + \tau(\mathbf{y} - \mathbf{x})$. By the Cauchy–Schwarz inequality, $\|\mathbf{u}\|^2 \leq \|\mathbf{u}\| \, \|D\mathbf{g}(\mathbf{z})\|_2 \, \|\mathbf{y} - \mathbf{x}\|$; divide by $\|\mathbf{u}\|$. □

Proposition A.5.2 (Theorem 3.6 of Lecture 3)**.** *If all partial derivatives of $f : U \to \mathbb{R}$ exist in a ball around $\mathbf{x}$ and are continuous at $\mathbf{x}$, then f is differentiable at $\mathbf{x}$.*

Proof. For small $\mathbf{h}$, let $\mathbf{z}_0 = \mathbf{x}$ and $\mathbf{z}_j = \mathbf{x} + h_1\mathbf{e}_1 + \cdots + h_j\mathbf{e}_j$, so that $\mathbf{z}_n = \mathbf{x} + \mathbf{h}$. The points $\mathbf{z}_{j-1}$ and $\mathbf{z}_j$ differ only in the j-th coordinate, so by the mean value theorem in one variable, $f(\mathbf{z}_j) - f(\mathbf{z}_{j-1}) = \frac{\partial f}{\partial x_j}(\boldsymbol{\xi}_j) h_j$ for a point $\boldsymbol{\xi}_j$ between them. Adding these equations,

$$\left| f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{h} \right| = \Big| \sum_{j=1}^{n} \Big(\frac{\partial f}{\partial x_j}(\boldsymbol{\xi}_j) - \frac{\partial f}{\partial x_j}(\mathbf{x}) \Big) h_j \Big| \leq \sum_{j=1}^{n} \Big| \frac{\partial f}{\partial x_j}(\boldsymbol{\xi}_j) - \frac{\partial f}{\partial x_j}(\mathbf{x}) \Big| \, \|\mathbf{h}\|.$$

As $\mathbf{h} \to \mathbf{0}$, all $\boldsymbol{\xi}_j \to \mathbf{x}$, so the sum tends to 0 by the continuity of the partial derivatives at $\mathbf{x}$. □

The key tool of this section is a simple but powerful theorem about solving equations by iteration.

Theorem A.5.3 (Banach fixed-point theorem)**.** *Let $C \subseteq \mathbb{R}^n$ be closed and nonempty, and let $T : C \to C$ be a contraction, i.e., $\|T(\mathbf{x}) - T(\mathbf{y})\| \le q\|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in C$, with a constant $q < 1$. Then T has exactly one fixed point $\mathbf{x}^* = T(\mathbf{x}^*)$. For every $\mathbf{x}_0 \in C$, the iterates $\mathbf{x}_{k+1} = T(\mathbf{x}_k)$ converge to $\mathbf{x}^*$, and*

$$\|\mathbf{x}_k - \mathbf{x}^*\| \le \frac{q^k}{1-q}\|\mathbf{x}_1 - \mathbf{x}_0\|.$$

Proof. By induction, $\|\mathbf{x}_{j+1} - \mathbf{x}_j\| \le q^j\|\mathbf{x}_1 - \mathbf{x}_0\|$. For $l > k$, the triangle inequality and the geometric series give

$$\|\mathbf{x}_l - \mathbf{x}_k\| \le \sum_{j=k}^{l-1} q^j\|\mathbf{x}_1 - \mathbf{x}_0\| \le \frac{q^k}{1-q}\|\mathbf{x}_1 - \mathbf{x}_0\|.$$

So $(\mathbf{x}_k)$ is a Cauchy sequence, which converges to some $\mathbf{x}^*$ by Corollary A.3.2, and $\mathbf{x}^* \in C$ since C is closed. Since T is continuous, $T(\mathbf{x}^*) = \lim T(\mathbf{x}_k) = \lim \mathbf{x}_{k+1} = \mathbf{x}^*$. Letting $l \to \infty$ in the inequality gives the error bound. If $\mathbf{y}^*$ is another fixed point, then $\|\mathbf{x}^* - \mathbf{y}^*\| = \|T(\mathbf{x}^*) - T(\mathbf{y}^*)\| \le q\|\mathbf{x}^* - \mathbf{y}^*\|$, so $\mathbf{x}^* = \mathbf{y}^*$. □

Example A.5.4. The function $T(x) = \cos x$ maps $[0, 1]$ into $[\cos 1, 1] \subseteq [0, 1]$, and $|T'(x)| = |\sin x| \le \sin 1 \approx 0.841$ there, so it is a contraction with $q = \sin 1$ (by the mean value theorem). So the equation $\cos x = x$ has exactly one solution in $[0, 1]$, $x^* = 0.739085\ldots$, and pressing the cosine key of a calculator repeatedly finds it. Starting at $x_0 = 1$, the error bound guarantees an accuracy of 10^{-6} after 87 steps; in fact, 32 steps suffice. Gradient descent on a quadratic function with $\rho < 1$ (Theorem 10.6 of Lecture 3) is another example of a contraction.

Corollary A.5.5 (Invertible residual layers)**.** *Let $\mathbf{g} : \mathbb{R}^n \to \mathbb{R}^n$ satisfy $\|\mathbf{g}(\mathbf{x}) - \mathbf{g}(\mathbf{y})\| \le q\|\mathbf{x} - \mathbf{y}\|$ with $q < 1$. Then the residual layer $\mathbf{f}(\mathbf{x}) = \mathbf{x} + \mathbf{g}(\mathbf{x})$ is a homeomorphism of $\mathbb{R}^n$, and $\mathbf{f}^{-1}(\mathbf{y})$ is the limit of the iteration $\mathbf{x}_{k+1} = \mathbf{y} - \mathbf{g}(\mathbf{x}_k)$.*

Proof. For fixed $\mathbf{y}$, the map $T(\mathbf{x}) = \mathbf{y} - \mathbf{g}(\mathbf{x})$ is a contraction of $\mathbb{R}^n$, and $\mathbf{f}(\mathbf{x}) = \mathbf{y}$ if and only if $\mathbf{x} = T(\mathbf{x})$. By Theorem A.5.3, there is exactly one solution, so $\mathbf{f}$ is bijective. It is continuous, and so is its inverse: $\|\mathbf{x} - \mathbf{x}'\| \le \|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}')\| + \|\mathbf{g}(\mathbf{x}) - \mathbf{g}(\mathbf{x}')\| \le \|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}')\| + q\|\mathbf{x} - \mathbf{x}'\|$, so $\|\mathbf{f}^{-1}(\mathbf{y}) - \mathbf{f}^{-1}(\mathbf{y}')\| \le \frac{1}{1-q}\|\mathbf{y} - \mathbf{y}'\|$. □

Invertible neural networks used in normalizing flows (Remark 4.8 of Lecture 3) are built from such layers: constraining the Lipschitz constant of each residual branch (e.g., by normalizing the spectral norms of its weight matrices) makes the whole network invertible. A diffeomorphism must have invertible Jacobians everywhere, by the chain rule applied to $\mathbf{f}^{-1} \circ \mathbf{f} = \mathrm{id}$. The inverse function theorem is a local converse.

Theorem A.5.6 (Inverse function theorem)**.** *Let $\mathbf{f} : U \to \mathbb{R}^n$ be continuously differentiable on an open set $U \subseteq \mathbb{R}^n$, and let $D\mathbf{f}(\mathbf{a})$ be invertible. Then there are open sets $V \ni \mathbf{a}$ and $W \ni \mathbf{f}(\mathbf{a})$ such that $\mathbf{f}$ maps V bijectively onto W, and the inverse $\mathbf{f}^{-1} : W \to V$ is continuously differentiable, with*

$$D(\mathbf{f}^{-1})(\mathbf{f}(\mathbf{x})) = D\mathbf{f}(\mathbf{x})^{-1} \qquad \text{for } \mathbf{x} \in V.$$

If $\mathbf{f}$ is smooth, so is $\mathbf{f}^{-1}$.

Sketch of proof. Replacing $\mathbf{f}(\mathbf{x})$ by $D\mathbf{f}(\mathbf{a})^{-1}(\mathbf{f}(\mathbf{a}+\mathbf{x})-\mathbf{f}(\mathbf{a}))$, we may assume $\mathbf{a}=\mathbf{0}$, $\mathbf{f}(\mathbf{0})=\mathbf{0}$ and $D\mathbf{f}(\mathbf{0})=I$. Then $\mathbf{g}(\mathbf{x})=\mathbf{x}-\mathbf{f}(\mathbf{x})$ has $D\mathbf{g}(\mathbf{0})=0$, so by continuity there is an $r>0$ such that the closed ball $\overline{B}_r=\{\|\mathbf{x}\|\le r\}$ lies in U and $\|D\mathbf{g}(\mathbf{x})\|_2\le\frac{1}{2}$ on it. By Lemma A.5.1, $\mathbf{g}$ is a contraction on $\overline{B}_r$ with $q=\frac{1}{2}$, and $\|\mathbf{g}(\mathbf{x})\|\le\frac{1}{2}\|\mathbf{x}\|$. For $\|\mathbf{y}\|<\frac{r}{2}$, the map $T(\mathbf{x})=\mathbf{y}+\mathbf{g}(\mathbf{x})$ maps $\overline{B}_r$ into itself, since $\|T(\mathbf{x})\|\le\frac{r}{2}+\frac{r}{2}$, and it is a contraction. Its unique fixed point is the unique $\mathbf{x}\in\overline{B}_r$ with $\mathbf{f}(\mathbf{x})=\mathbf{y}$, and it satisfies $\|\mathbf{x}\|\le\|\mathbf{y}\|+\frac{1}{2}\|\mathbf{x}\|$, i.e., $\|\mathbf{x}\|\le 2\|\mathbf{y}\|<r$. So with $W=B(\mathbf{0},\frac{r}{2})$ and $V=B(\mathbf{0},r)\cap\mathbf{f}^{-1}(W)$, $\mathbf{f}:V\to W$ is bijective, and as in Corollary A.5.5, the inverse is continuous. One can show that it is also differentiable; then the chain rule applied to $\mathbf{f}^{-1}(\mathbf{f}(\mathbf{x}))=\mathbf{x}$ gives $D(\mathbf{f}^{-1})(\mathbf{f}(\mathbf{x}))\,D\mathbf{f}(\mathbf{x})=I$. □

Example A.5.7. 1. Polar coordinates $\mathbf{f}(r,\varphi)=(r\cos\varphi, r\sin\varphi)$ have $\det D\mathbf{f}=r$ (Example 4.7 of Lecture 3). So near every point with $r>0$, they have a smooth local inverse. But $\mathbf{f}$ is not injective, since $\mathbf{f}(r,\varphi+2\pi)=\mathbf{f}(r,\varphi)$: the theorem is only local.

2. The map $F(x,y)=(x^2-y^2,2xy)$, i.e., $z\mapsto z^2$ for $z=x+iy$, has $\det DF=4(x^2+y^2)$. Near every point $z\neq 0$, it has a local inverse, a "branch of the square root". Globally, it is two-to-one, since $F(-z)=F(z)$, and at $z=0$, it is not even locally injective.

A.6 The Implicit Function Theorem

When can an equation $F(\mathbf{x},\mathbf{y})=\mathbf{0}$ be solved for some of the variables? We write points of $\mathbb{R}^{k+m}$ as $(\mathbf{x},\mathbf{y})$ with $\mathbf{x}\in\mathbb{R}^k$ and $\mathbf{y}\in\mathbb{R}^m$, and split the Jacobian of $F:\mathbb{R}^{k+m}\to\mathbb{R}^m$ into the $m\times k$ matrix $D_{\mathbf{x}}F$ and the square $m\times m$ matrix $D_{\mathbf{y}}F$ of partial derivatives with respect to $\mathbf{x}$ and $\mathbf{y}$.

Theorem A.6.1 (Implicit function theorem)**.** *Let $F:U\to\mathbb{R}^m$ be continuously differentiable on an open set $U\subseteq\mathbb{R}^{k+m}$, let $F(\mathbf{a},\mathbf{b})=\mathbf{0}$, and let $D_{\mathbf{y}}F(\mathbf{a},\mathbf{b})$ be invertible. Then there are open sets $V\ni\mathbf{a}$ and $W\ni(\mathbf{a},\mathbf{b})$ and a continuously differentiable function $\mathbf{g}:V\to\mathbb{R}^m$ with $\mathbf{g}(\mathbf{a})=\mathbf{b}$, such that for $(\mathbf{x},\mathbf{y})\in W$,*

$$F(\mathbf{x},\mathbf{y})=\mathbf{0}\quad\Longleftrightarrow\quad\mathbf{x}\in V\text{ and }\mathbf{y}=\mathbf{g}(\mathbf{x}).$$

Moreover, $D\mathbf{g}(\mathbf{x})=-D_{\mathbf{y}}F(\mathbf{x},\mathbf{g}(\mathbf{x}))^{-1}\,D_{\mathbf{x}}F(\mathbf{x},\mathbf{g}(\mathbf{x}))$.

Sketch of proof. Let $\Phi(\mathbf{x},\mathbf{y})=(\mathbf{x},F(\mathbf{x},\mathbf{y}))$. Its Jacobian $\begin{pmatrix} I & 0\\ D_{\mathbf{x}}F & D_{\mathbf{y}}F\end{pmatrix}$ is invertible at $(\mathbf{a},\mathbf{b})$: the equations $\mathbf{h}=\mathbf{u}$, $D_{\mathbf{x}}F\,\mathbf{h}+D_{\mathbf{y}}F\,\mathbf{k}=\mathbf{v}$ have the unique solution $\mathbf{h}=\mathbf{u}$, $\mathbf{k}=D_{\mathbf{y}}F^{-1}(\mathbf{v}-D_{\mathbf{x}}F\,\mathbf{u})$. By Theorem A.5.6, Φ has a local inverse near $\Phi(\mathbf{a},\mathbf{b})=(\mathbf{a},\mathbf{0})$. Since Φ does not change $\mathbf{x}$, neither does its inverse, which therefore has the form $(\mathbf{x},\mathbf{z})\mapsto(\mathbf{x},G(\mathbf{x},\mathbf{z}))$. Near $(\mathbf{a},\mathbf{b})$, $F(\mathbf{x},\mathbf{y})=\mathbf{0}$ means $\Phi(\mathbf{x},\mathbf{y})=(\mathbf{x},\mathbf{0})$, i.e., $\mathbf{y}=G(\mathbf{x},\mathbf{0})=:\mathbf{g}(\mathbf{x})$. Differentiating $F(\mathbf{x},\mathbf{g}(\mathbf{x}))=\mathbf{0}$ with the chain rule gives $D_{\mathbf{x}}F+D_{\mathbf{y}}F\,D\mathbf{g}=0$. □

Example A.6.2. 1. The unit circle $F(x,y)=x^2+y^2-1=0$ has $\frac{\partial F}{\partial y}=2y$. Near every point with $y\neq 0$, the circle is the graph of a function of x ($y=\pm\sqrt{1-x^2}$), with $\frac{dy}{dx}=-\frac{2x}{2y}$. Near $(\pm1,0)$, it is not, but it is the graph of a function of y, since $\frac{\partial F}{\partial x}=2x\neq 0$ there.

2. For the folium of Descartes $x^3+y^3=3xy$ (Example 9.4 of Lecture 3), $\frac{\partial F}{\partial y}=3y^2-3x$ vanishes on the curve only at $(0,0)$, where the curve crosses itself, and at $(2^{2/3},2^{1/3})$, where its tangent is vertical. Near every other point of the curve, y is a smooth function of x.

Remark A.6.3 (Implicit differentiation in machine learning). Sometimes a quantity is defined as the solution of an equation, $G(\mathbf{z}, \boldsymbol{\theta}) = \mathbf{0}$, and we need its derivative with respect to $\boldsymbol{\theta}$. Examples are the minimizer $\mathbf{z}^*(\boldsymbol{\theta})$ of an inner optimization problem with hyperparameters $\boldsymbol{\theta}$ (here $G = \nabla_{\mathbf{z}}$ of the inner objective), and the fixed point $\mathbf{z} = \mathbf{h}(\mathbf{z}, \boldsymbol{\theta})$ of an "implicit layer" in deep equilibrium models. By Theorem A.6.1, $D\mathbf{z}^*(\boldsymbol{\theta}) = -D_{\mathbf{z}}G^{-1}D_{\boldsymbol{\theta}}G$, so we do not need to backpropagate through all the iterations of the solver that computed $\mathbf{z}^*$. For example, the ridge regression solution $\mathbf{w}^*(\lambda)$ solves $G(\mathbf{w}, \lambda) = X^\top(X\mathbf{w} - \mathbf{y}) + \lambda\mathbf{w} = \mathbf{0}$, so

$$\frac{d\mathbf{w}^*}{d\lambda} = -(X^\top X + \lambda I)^{-1}\mathbf{w}^*(\lambda),$$

which can be used to tune λ by gradient descent on a validation loss.

A.7 Submanifolds and Tangent Spaces

A k-dimensional manifold is a set that looks like $\mathbb{R}^k$ near each of its points: a curve looks like a line, and a surface like a plane. We consider manifolds that lie in $\mathbb{R}^n$ and are described by equations.

Definition A.7.1 (Submanifold)**.** A set $M \subseteq \mathbb{R}^n$ is a k-dimensional submanifold of $\mathbb{R}^n$ if for every $\mathbf{p} \in M$, there are an open set $U \ni \mathbf{p}$ and a smooth map $F : U \to \mathbb{R}^{n-k}$ such that

$$M \cap U = \{\mathbf{x} \in U : F(\mathbf{x}) = \mathbf{0}\} \qquad \text{and} \qquad \operatorname{rank} DF(\mathbf{p}) = n - k.$$

(For $k = n$, there is no equation: M is an open subset of $\mathbb{R}^n$.)

So locally, M is cut out by $n-k$ independent equations. In particular, if $F : U \to \mathbb{R}^m$ is smooth and $\mathbf{c} \in \mathbb{R}^m$ is a regular value of F, i.e., $\operatorname{rank} DF(\mathbf{x}) = m$ for every $\mathbf{x}$ with $F(\mathbf{x}) = \mathbf{c}$, then the level set $F^{-1}(\mathbf{c})$ is an $(n-m)$-dimensional submanifold (use $F - \mathbf{c}$ in the definition). The implicit function theorem shows that submanifolds really look like $\mathbb{R}^k$.

Proposition A.7.2 (Submanifolds are local graphs)**.** *Let M be a k-dimensional submanifold and $\mathbf{p} \in M$. After renumbering the coordinates of $\mathbb{R}^n = \mathbb{R}^k \times \mathbb{R}^{n-k}$, there are open sets $V \subseteq \mathbb{R}^k$ and $W \ni \mathbf{p}$ and a smooth map $\mathbf{g} : V \to \mathbb{R}^{n-k}$ such that $M \cap W = \{(\mathbf{x}, \mathbf{g}(\mathbf{x})) : \mathbf{x} \in V\}$. In particular, $\mathbf{x} \mapsto (\mathbf{x}, \mathbf{g}(\mathbf{x}))$ is a homeomorphism from the open set $V \subseteq \mathbb{R}^k$ onto $M \cap W$.*

Proof. The $(n-k) \times n$ matrix $DF(\mathbf{p})$ has rank $n-k$, so it has $n-k$ linearly independent columns (Lecture 1). Renumber the coordinates so that these are the last $n-k$ columns, and write $\mathbf{x} \in \mathbb{R}^n$ as $(\mathbf{x}', \mathbf{y})$. Then $D_{\mathbf{y}}F(\mathbf{p})$ is invertible, and Theorem A.6.1 gives $\mathbf{g}$. The inverse of $\mathbf{x}' \mapsto (\mathbf{x}', \mathbf{g}(\mathbf{x}'))$ is the projection to the first k coordinates, which is continuous. □

Example A.7.3.

1. The sphere $S^{n-1} = \{\mathbf{x} : \|\mathbf{x}\|^2 = 1\}$ is an $(n-1)$-dimensional submanifold, since $F(\mathbf{x}) = \|\mathbf{x}\|^2 - 1$ has $DF(\mathbf{x}) = 2\mathbf{x}^\top \neq \mathbf{0}$ on the sphere.
2. The graph $\{(\mathbf{x}, \mathbf{g}(\mathbf{x}))\}$ of a smooth $\mathbf{g} : \mathbb{R}^k \to \mathbb{R}^{n-k}$ is a k-dimensional submanifold: $F(\mathbf{x}, \mathbf{y}) = \mathbf{y} - \mathbf{g}(\mathbf{x})$ has $DF = (-D\mathbf{g},\, I)$, of rank $n-k$.
3. $\mathrm{GL}(n, \mathbb{R})$ is open in $\mathbb{R}^{n \times n}$, hence an n^2-dimensional submanifold.
4. The special linear group $\mathrm{SL}(n, \mathbb{R}) = \{X : \det X = 1\}$ is a submanifold of dimension $n^2 - 1$. By Jacobi's formula (Proposition 6.4 of Lecture 3), $D\det(X)H = \det(X)\operatorname{tr}(X^{-1}H)$, which for $\det X = 1$ and $H = X$ equals $n \neq 0$. So $D\det(X) \neq 0$, and 1 is a regular value of det.

5. The orthogonal group $\mathrm{O}(n) = \{Q : Q^\top Q = I\}$ is a submanifold of dimension $\frac{n(n-1)}{2}$. Let $F(Q) = Q^\top Q$, a map from $\mathbb{R}^{n\times n}$ to the space Sym_n of symmetric matrices, which has dimension $\frac{n(n+1)}{2}$. Its derivative is $DF(Q)H = H^\top Q + Q^\top H$. For $Q \in \mathrm{O}(n)$ and any symmetric S, the matrix $H = \frac{1}{2}QS$ satisfies $DF(Q)H = \frac{1}{2}SQ^\top Q + \frac{1}{2}Q^\top QS = S$, so $DF(Q)$ is surjective, i.e., has full rank $\frac{n(n+1)}{2}$, and $\mathrm{O}(n)$ has dimension $n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$. For $n = 2$, this is 1 (one rotation angle), and for $n = 3$, it is 3 (three angles, as for the rotations R_1, R_2, R_3 of Lecture 2).

Not every level set is a submanifold. The level set $\{x^2 - y^2 = 0\}$ of Figure 1 in Lecture 3 consists of two lines that cross at the origin, where the gradient $(2x, -2y)$ vanishes. It is not a 1-dimensional submanifold near the origin: removing the origin from a small piece of it leaves four connected pieces, while removing a point from a small piece of a curve that looks like a line leaves two. Similarly, the double cone $\{x^2 + y^2 = z^2\}$ is not a submanifold at its tip. On the other hand, the rank condition is sufficient, but not necessary: $\{x^2 = 0\}$ is the y-axis, a perfectly good submanifold, although $D(x^2) = (2x, 0)$ vanishes on it.

Definition A.7.4 (Tangent space)**.** Let M be a submanifold and $\mathbf{p} \in M$. The tangent space $T_\mathbf{p}M$ is the set of velocities $\boldsymbol{\gamma}'(0)$ of all smooth curves $\boldsymbol{\gamma} : (-\varepsilon, \varepsilon) \to \mathbb{R}^n$ with $\boldsymbol{\gamma}(t) \in M$ for all t and $\boldsymbol{\gamma}(0) = \mathbf{p}$. Its orthogonal complement $N_\mathbf{p}M = (T_\mathbf{p}M)^\perp$ is the normal space.

Theorem A.7.5. *Let M be a k-dimensional submanifold, and let $M \cap U = F^{-1}(\mathbf{0})$ near $\mathbf{p}$ as in Definition A.7.1. Then*

$$T_\mathbf{p}M = \operatorname{null} DF(\mathbf{p}) \qquad \text{and} \qquad N_\mathbf{p}M = \operatorname{row} DF(\mathbf{p}) = \operatorname{span}\{\nabla F_1(\mathbf{p}), \dots, \nabla F_{n-k}(\mathbf{p})\}.$$

In particular, $T_\mathbf{p}M$ is a k-dimensional subspace.

Proof. If $\boldsymbol{\gamma}$ is a curve in M through $\mathbf{p}$, then $F(\boldsymbol{\gamma}(t)) = \mathbf{0}$ for small t, and the chain rule gives $DF(\mathbf{p})\boldsymbol{\gamma}'(0) = \mathbf{0}$. So $T_\mathbf{p}M \subseteq \operatorname{null} DF(\mathbf{p})$, which has dimension $n - (n-k) = k$ by the rank–nullity theorem (Lecture 1). Conversely, write M near $\mathbf{p} = (\mathbf{a}, \mathbf{g}(\mathbf{a}))$ as a graph as in Proposition A.7.2. For every $\mathbf{v} \in \mathbb{R}^k$, the curve $t \mapsto (\mathbf{a} + t\mathbf{v}, \mathbf{g}(\mathbf{a} + t\mathbf{v}))$ lies in M and has velocity $(\mathbf{v}, D\mathbf{g}(\mathbf{a})\mathbf{v})$ at $t = 0$. These velocities form a k-dimensional subspace (the image of an injective linear map), so $T_\mathbf{p}M$ contains a k-dimensional subspace of the k-dimensional space $\operatorname{null} DF(\mathbf{p})$, and the two are equal. Finally, $\operatorname{null}(A)^\perp = \operatorname{row}(A)$ for every matrix A (Lecture 2), and the rows of $DF(\mathbf{p})$ are the transposed gradients $\nabla F_i(\mathbf{p})^\top$. □

Example A.7.6.

1. $T_\mathbf{p}S^{n-1} = \operatorname{null}(2\mathbf{p}^\top) = \{\mathbf{v} : \mathbf{p}^\top\mathbf{v} = 0\} = \mathbf{p}^\perp$: the tangent vectors of a sphere are orthogonal to the radius (Figure 1).

2. $T_I\mathrm{SL}(n, \mathbb{R}) = \{H : D\det(I)H = 0\} = \{H : \operatorname{tr} H = 0\}$, the traceless matrices: to first order, $\det(I + tH) = 1 + t\operatorname{tr} H$ (Lemma 6.3 of Lecture 3), so a matrix curve through I keeps determinant 1 only if its velocity has trace 0.

3. $T_Q\mathrm{O}(n) = \{H : H^\top Q + Q^\top H = 0\}$. With $K = Q^\top H$, the condition says $K^\top + K = 0$, so $T_Q\mathrm{O}(n) = \{QK : K^\top = -K\}$, and $T_I\mathrm{O}(n)$ is the space of skew-symmetric matrices, of dimension $\frac{n(n-1)}{2}$. For example, the rotations $R(t)$ by the angle t in the plane have $R'(0) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

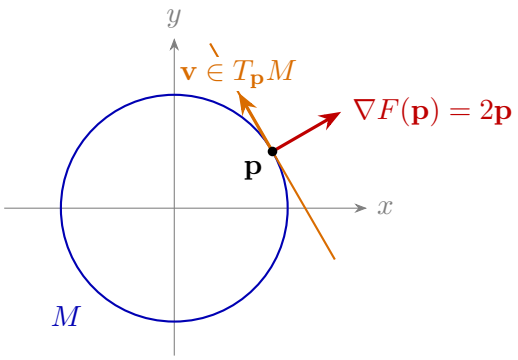


Figure 1: The circle $M = \{F = 0\}$ with $F(x, y) = x^2 + y^2 - 1$. The tangent space $T_{\mathbf{p}}M = \operatorname{null} DF(\mathbf{p}) = \mathbf{p}^{\perp}$ is the tangent line (shifted to $\mathbf{p}$), and the normal space is spanned by the gradient $\nabla F(\mathbf{p})$.

A.8 Lagrange Multipliers and Optimization on Manifolds

We can now complete the proof of the Lagrange multiplier theorem (Theorem 12.1 of Lecture 3).

Theorem A.8.1 (Lagrange multipliers)**.** *Let f and $g_1, \dots, g_m$ be smooth on an open set $U \subseteq \mathbb{R}^n$, let $\mathbf{x}^*$ be a local minimum of f on $M = \{\mathbf{x} \in U : g_1(\mathbf{x}) = \cdots = g_m(\mathbf{x}) = 0\}$, and let $\nabla g_1(\mathbf{x}^*), \dots, \nabla g_m(\mathbf{x}^*)$ be linearly independent. Then $\nabla f(\mathbf{x}^*) \in \operatorname{span}\{\nabla g_1(\mathbf{x}^*), \dots, \nabla g_m(\mathbf{x}^*)\}$, i.e., there are λ_i with $\nabla f(\mathbf{x}^*) + \sum_i \lambda_i \nabla g_i(\mathbf{x}^*) = \mathbf{0}$.*

Proof. The linear independence says that $DG(\mathbf{x}^*)$ has rank m for $G = (g_1, \dots, g_m)$, so near $\mathbf{x}^*$, M is an $(n-m)$-dimensional submanifold (the rank stays m near $\mathbf{x}^*$, since a nonzero $m \times m$ minor stays nonzero by continuity). For every $\mathbf{v} \in T_{\mathbf{x}^*}M$, choose a curve $\boldsymbol{\gamma}$ in M with $\boldsymbol{\gamma}(0) = \mathbf{x}^*$ and $\boldsymbol{\gamma}'(0) = \mathbf{v}$. The function $t \mapsto f(\boldsymbol{\gamma}(t))$ has a local minimum at $t = 0$, so its derivative $\nabla f(\mathbf{x}^*)^{\top}\mathbf{v}$ is 0. Hence $\nabla f(\mathbf{x}^*) \in (T_{\mathbf{x}^*}M)^{\perp} = N_{\mathbf{x}^*}M = \operatorname{span}\{\nabla g_i(\mathbf{x}^*)\}$ by Theorem A.7.5. □

The same method proves the necessity of the KKT conditions for inequality constraints (Theorem 12.5 of Lecture 3). The new point is the sign of the multipliers: we move from $\mathbf{x}^*$ into the feasible region along a curve on which one active constraint becomes negative and the other active constraints stay 0.

Theorem A.8.2 (KKT conditions)**.** *Let $f, g_1, \dots, g_m$ be smooth on an open set $U \subseteq \mathbb{R}^n$, and let $\mathbf{x}^*$ be a local minimum of f on the feasible set $\{\mathbf{x} \in U : g_1(\mathbf{x}) \le 0, \dots, g_m(\mathbf{x}) \le 0\}$, such that the gradients $\nabla g_i(\mathbf{x}^*)$ of the active constraints, i.e., of those with $g_i(\mathbf{x}^*) = 0$, are linearly independent. Then there are numbers $\mu_1, \dots, \mu_m \ge 0$ such that $\nabla f(\mathbf{x}^*) + \sum_i \mu_i \nabla g_i(\mathbf{x}^*) = \mathbf{0}$, and $\mu_i = 0$ for every inactive constraint.*

Proof. Let A be the set of the indices of the active constraints. The inactive constraints satisfy $g_i(\mathbf{x}^*) < 0$, so by continuity, there is a ball $W \subseteq U$ around $\mathbf{x}^*$ on which $g_i < 0$ for all $i \notin A$, and we may choose it so small that $f(\mathbf{x}) \ge f(\mathbf{x}^*)$ for all feasible $\mathbf{x} \in W$. We set $\mu_i = 0$ for $i \notin A$. If A is empty, then all points of W are feasible, so $\mathbf{x}^*$ is a local minimum of f on an open set, and $\nabla f(\mathbf{x}^*) = \mathbf{0}$. Otherwise, all points of $M = \{\mathbf{x} \in W : g_i(\mathbf{x}) = 0 \text{ for all } i \in A\}$ are feasible, so $\mathbf{x}^*$ is a local minimum of f on M, and Theorem A.8.1 gives numbers μ_i, $i \in A$, with $\nabla f(\mathbf{x}^*) + \sum_{i \in A} \mu_i \nabla g_i(\mathbf{x}^*) = \mathbf{0}$.

It remains to show that $\mu_j \ge 0$ for every $j \in A$. The matrix with the rows $\nabla g_i(\mathbf{x}^*)^{\top}$, $i \in A$, has linearly independent rows, so its column space is all of $\mathbb{R}^{|A|}$ (Lecture 1), and there is a vector

$\mathbf{v}$ with

$$\nabla g_j(\mathbf{x}^*)^\top \mathbf{v} = -1 \qquad \text{and} \qquad \nabla g_i(\mathbf{x}^*)^\top \mathbf{v} = 0 \quad \text{for } i \in A,\ i \neq j.$$

If $A = \{j\}$, let $\boldsymbol{\gamma}(t) = \mathbf{x}^* + t\mathbf{v}$. Otherwise, the gradients of the constraints g_i, $i \in A \setminus \{j\}$, are linearly independent, so after shrinking W, $M_j = \{\mathbf{x} \in W : g_i(\mathbf{x}) = 0 \text{ for } i \in A \setminus \{j\}\}$ is a submanifold, as in the proof of Theorem A.8.1. By Theorem A.7.5, $\mathbf{v} \in T_{\mathbf{x}^*} M_j$, so there is a smooth curve $\boldsymbol{\gamma}$ in M_j with $\boldsymbol{\gamma}(0) = \mathbf{x}^*$ and $\boldsymbol{\gamma}'(0) = \mathbf{v}$. In both cases, the constraints g_i with $i \in A \setminus \{j\}$ are 0 along $\boldsymbol{\gamma}$, and the derivative of $g_j(\boldsymbol{\gamma}(t))$ at $t = 0$ is $\nabla g_j(\mathbf{x}^*)^\top \mathbf{v} = -1 < 0$, so $g_j(\boldsymbol{\gamma}(t)) < g_j(\mathbf{x}^*) = 0$ for small $t > 0$. Hence $\boldsymbol{\gamma}(t)$ is feasible for small $t > 0$, and $f(\boldsymbol{\gamma}(t)) \geq f(\mathbf{x}^*)$. So the difference quotients of $f(\boldsymbol{\gamma}(t))$ at $t = 0$ are nonnegative for $t > 0$, and so is its derivative:

$$0 \leq \nabla f(\mathbf{x}^*)^\top \mathbf{v} = -\sum_{i \in A} \mu_i \nabla g_i(\mathbf{x}^*)^\top \mathbf{v} = \mu_j. \qquad \square$$

The proof suggests how to do calculus on a manifold: only the component of ∇f in the tangent space matters.

Definition A.8.3 (Riemannian gradient)**.** Let M be a submanifold of $\mathbb{R}^n$, $\mathbf{p} \in M$, and f smooth near M. The Riemannian gradient of f on M at $\mathbf{p}$ is the orthogonal projection (Lecture 2) of $\nabla f(\mathbf{p})$ onto $T_{\mathbf{p}} M$:

$$\operatorname{grad}_M f(\mathbf{p}) = P_{\mathbf{p}} \nabla f(\mathbf{p}).$$

Since $\nabla f(\mathbf{p}) - \operatorname{grad}_M f(\mathbf{p})$ is orthogonal to $T_{\mathbf{p}} M$, we have $\nabla f(\mathbf{p})^\top \mathbf{v} = \operatorname{grad}_M f(\mathbf{p})^\top \mathbf{v}$ for all tangent vectors $\mathbf{v}$: the Riemannian gradient gives the rates of change of f along M. As in Theorem 3.9 of Lecture 3, it points in the direction of steepest ascent among the tangent directions, and by Theorem A.8.1, $\operatorname{grad}_M f(\mathbf{p}) = \mathbf{0}$ exactly at the points that satisfy the Lagrange condition. These points are the critical points of f on M.

Riemannian gradient descent minimizes f on M by taking a step in the direction $-\operatorname{grad}_M f(\mathbf{p}_k)$, which leaves M slightly, and mapping the result back to M with a retraction R:

$$\mathbf{p}_{k+1} = R(\mathbf{p}_k - \eta \operatorname{grad}_M f(\mathbf{p}_k)).$$

On the sphere, $P_{\mathbf{p}} = I - \mathbf{p}\mathbf{p}^\top$, so $\operatorname{grad}_M f(\mathbf{p}) = \nabla f(\mathbf{p}) - (\mathbf{p}^\top \nabla f(\mathbf{p}))\mathbf{p}$, and the retraction can simply normalize: $R(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|}$. On $\mathrm{O}(n)$, one can retract with the Q factor of the QR decomposition (Lecture 2). Such methods are used for PCA, for orthogonality constraints on weight matrices, and for low-rank matrix problems.

Example A.8.4 (The Rayleigh quotient on the sphere)**.** Let A be symmetric with eigenvalues $\lambda_1 > \lambda_2 > \cdots > \lambda_n$ and unit eigenvectors $\mathbf{q}_1, \ldots, \mathbf{q}_n$, and let $f(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}$ on S^{n-1}. Then $\operatorname{grad}_M f(\mathbf{x}) = 2A\mathbf{x} - 2(\mathbf{x}^\top A \mathbf{x})\mathbf{x}$, which is $\mathbf{0}$ if and only if $\mathbf{x}$ is an eigenvector: the critical points are the $2n$ points $\pm\mathbf{q}_i$. To see their types, move from $\mathbf{q}_i$ towards $\mathbf{q}_j$ along the great circle $\boldsymbol{\gamma}(t) = \cos t\, \mathbf{q}_i + \sin t\, \mathbf{q}_j$:

$$f(\boldsymbol{\gamma}(t)) = \lambda_i \cos^2 t + \lambda_j \sin^2 t = \lambda_i + (\lambda_j - \lambda_i) \sin^2 t.$$

So f decreases in the $n - i$ tangent directions $\mathbf{q}_j$ with $\lambda_j < \lambda_i$, and increases in the $i - 1$ directions with $\lambda_j > \lambda_i$. Hence $\pm\mathbf{q}_1$ are the maxima, $\pm\mathbf{q}_n$ the minima, and all other eigenvectors are saddle points. In particular, Riemannian gradient ascent for the first principal component (Lecture 2) has exactly one pair of local maxima, the correct answer; the other critical points are saddle points, which gradient ascent avoids from almost every starting point.

A.9 Critical Points and Morse Theory

A.9.1 Regular and critical values

Definition A.9.1 (Critical points and values of maps)**.** Let $F : U \to \mathbb{R}^m$ be smooth on an open set $U \subseteq \mathbb{R}^n$. A point $\mathbf{x}$ is a critical point of F if $\operatorname{rank} DF(\mathbf{x}) < m$, and a value $\mathbf{c}$ is a critical value if $\mathbf{c} = F(\mathbf{x})$ for a critical point $\mathbf{x}$. All other values in $\mathbb{R}^m$ are regular values.

For $m = 1$, the critical points are the points with $\nabla F(\mathbf{x}) = \mathbf{0}$, as in Lecture 3. By Section A.7, the level set of a regular value is a submanifold (or empty). How many critical values can there be? A set $Z \subseteq \mathbb{R}^m$ has measure zero if for every $\varepsilon > 0$ it can be covered by countably many boxes whose volumes add up to less than ε. For example, finite and countable sets, and lines in $\mathbb{R}^2$, have measure zero.

Theorem A.9.2 (Sard)**.** *The set of critical values of a smooth map $F : U \to \mathbb{R}^m$ has measure zero.*

We do not prove this theorem. It says that almost every value is regular, so almost every level set of a smooth function $f : U \to \mathbb{R}$ is an $(n-1)$-dimensional submanifold or empty. Note that the theorem is about values, not points: a constant function has only critical points, but only one critical value.

A.9.2 Nondegenerate critical points

From now on, let $f : U \to \mathbb{R}$ be smooth. Lecture 3 classified critical points by the eigenvalues of the Hessian (Theorem 9.3). The test was inconclusive when an eigenvalue is 0.

Definition A.9.3 (Nondegenerate critical points and the index)**.** A critical point $\mathbf{p}$ of f is nondegenerate if the Hessian $\nabla^2 f(\mathbf{p})$ is invertible, i.e., has no eigenvalue 0. Its index is the number of negative eigenvalues of $\nabla^2 f(\mathbf{p})$, counted with multiplicity. A function whose critical points are all nondegenerate is a Morse function.

So a nondegenerate critical point of index 0 is a strict local minimum, one of index n is a strict local maximum, and all others are saddle points. The index counts the independent directions in which f decreases. For example, $x^3 + y^3 - 3xy$ (Example 9.4 of Lecture 3) is a Morse function, with a critical point of index 1 at the origin and one of index 0 at $(1, 1)$. The "monkey saddle" $x^3 - 3xy^2$ has a degenerate critical point at the origin, where its Hessian is 0; near the origin, it goes up and down three times, unlike any nondegenerate saddle.

Proposition A.9.4. *Nondegenerate critical points are isolated: each one has a neighborhood that contains no other critical point.*

Proof. Let $H = \nabla^2 f(\mathbf{p})$, and let $\mu > 0$ be the smallest absolute value of an eigenvalue of H. Writing $\mathbf{h}$ in an orthonormal eigenbasis (Lecture 2) shows $\|H\mathbf{h}\| \geq \mu\|\mathbf{h}\|$. The gradient is differentiable with derivative H at $\mathbf{p}$, so $\nabla f(\mathbf{p} + \mathbf{h}) = H\mathbf{h} + \mathbf{r}(\mathbf{h})$ with $\|\mathbf{r}(\mathbf{h})\| \leq \frac{\mu}{2}\|\mathbf{h}\|$ for small $\mathbf{h}$. Then $\|\nabla f(\mathbf{p} + \mathbf{h})\| \geq \frac{\mu}{2}\|\mathbf{h}\| > 0$ for small $\mathbf{h} \neq \mathbf{0}$. □

In Lecture 3, we saw that near a critical point, f is approximately a sum of parabolas, $f(\mathbf{p}) + \frac{1}{2}\sum_i \lambda_i y_i^2$, in the coordinates given by the eigenvectors of the Hessian. For nondegenerate critical points, this becomes exact after a nonlinear change of coordinates.

Theorem A.9.5 (Morse lemma)**.** *Let* $\mathbf{p}$ *be a nondegenerate critical point of* f *with index* k*. Then there is a diffeomorphism* $\boldsymbol{\varphi}$ *from a neighborhood of* $\mathbf{p}$ *onto a neighborhood of* $\mathbf{0}$*, with* $\boldsymbol{\varphi}(\mathbf{p}) = \mathbf{0}$*, such that in the coordinates* $\mathbf{y} = \boldsymbol{\varphi}(\mathbf{x})$*,*

$$f = f(\mathbf{p}) - y_1^2 - \cdots - y_k^2 + y_{k+1}^2 + \cdots + y_n^2.$$

We do not prove the Morse lemma. It says that near a nondegenerate critical point, the higher-order terms of the Taylor expansion do not change the picture at all: up to a smooth deformation, the function is a bowl, a saddle with k downward directions, or an upside-down bowl. Degenerate critical points can be much more complicated, but they are rare.

Theorem A.9.6 (Morse functions are generic)**.** *Let* $f : U \to \mathbb{R}$ *be smooth. For almost every* $\mathbf{a} \in \mathbb{R}^n$ *(all except a set of measure zero), the function* $f_{\mathbf{a}}(\mathbf{x}) = f(\mathbf{x}) - \mathbf{a}^\top \mathbf{x}$ *is a Morse function.*

Proof. We have $\nabla f_{\mathbf{a}} = \nabla f - \mathbf{a}$ and $\nabla^2 f_{\mathbf{a}} = \nabla^2 f$. So $\mathbf{x}$ is a critical point of $f_{\mathbf{a}}$ if and only if $\nabla f(\mathbf{x}) = \mathbf{a}$, and it is degenerate if and only if $\nabla^2 f(\mathbf{x})$, the Jacobian of the map $\nabla f : U \to \mathbb{R}^n$, is singular. Hence $f_{\mathbf{a}}$ has a degenerate critical point if and only if $\mathbf{a}$ is a critical value of ∇f, and by Sard's theorem, these $\mathbf{a}$ form a set of measure zero. □

For example, $f(x) = x^3$ has a degenerate critical point at 0, but $x^3 - ax$ is a Morse function for every $a \neq 0$: it has two nondegenerate critical points if $a > 0$, and none if $a < 0$. An arbitrarily small, random linear perturbation removes all degenerate critical points.

A.9.3 Critical points and the shape of sublevel sets

Morse theory studies how the sublevel sets $M^c = \{\mathbf{x} : f(\mathbf{x}) \leq c\}$ change as c increases. We state its two main results informally, for a Morse function f whose sublevel sets are compact (for example, a Morse function on a compact manifold, or a coercive Morse function on $\mathbb{R}^n$).

- If there are no critical values in $[a, b]$, then M^a and M^b have the same shape: M^b can be continuously deformed into M^a, by flowing downhill along $-\nabla f$. Nothing interesting happens between critical values.
- When c passes the value of a critical point of index k, the sublevel set changes by attaching a k-dimensional "cell" (a point for $k = 0$, a segment for $k = 1$, a disk for $k = 2$, and so on). A minimum ($k = 0$) creates a new connected component. A saddle point of index 1 attaches a bridge, which either connects two components or closes a loop.

Example A.9.7 (A double well)**.** The function $f(x, y) = (x^2 - 1)^2 + y^2$ has $\nabla f = (4x(x^2 - 1), 2y)$, so its critical points are $(\pm 1, 0)$, with $f = 0$ and $\nabla^2 f = \operatorname{diag}(8, 2)$ (index 0), and $(0, 0)$, with $f = 1$ and $\nabla^2 f = \operatorname{diag}(-4, 2)$ (index 1). For $c < 0$, the sublevel set is empty; at $c = 0$, two components appear at the minima; at $c = 1$, the saddle point attaches a bridge that joins them, and for $c > 1$, the sublevel set is connected (Figure 2). For gradient descent, this means that the two basins of the minima are separated by the saddle: an iteration that starts with $f(\mathbf{x}_0) < 1$ and decreases f in every step cannot move from one basin to the other.

On a compact manifold, these changes must add up. The Euler characteristic $\chi(M)$ is a number that only depends on the topology of M: for a surface divided into V vertices, E edges and F faces, it is $V - E + F$, which does not depend on the subdivision. For example, the sphere has $\chi = 2$ (a cube has $8 - 12 + 6 = 2$), and the torus (the surface of a doughnut) has $\chi = 0$.

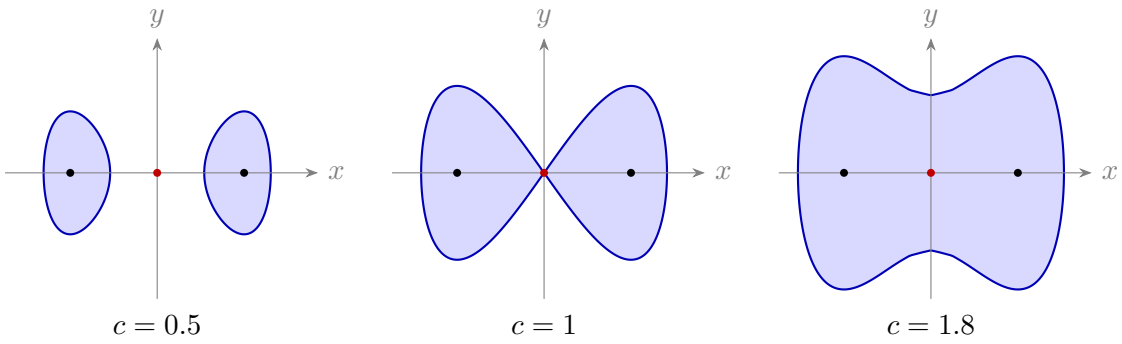


Figure 2: Sublevel sets $\{f \le c\}$ of the double well $f(x,y) = (x^2-1)^2 + y^2$, with the two minima (black, index 0, value 0) and the saddle point (red, index 1, value 1). Below the critical value 1, the sublevel set has two components; at $c = 1$, they touch at the saddle point; above it, they have merged into one.

Theorem A.9.8 (Morse relation)**.** *Let f be a Morse function on a compact manifold M, and let c_k be the number of its critical points of index k. Then*

$$\chi(M) = c_0 - c_1 + c_2 - \cdots + (-1)^n c_n.$$

We do not prove this theorem; it follows from the fact that attaching a k-cell changes the Euler characteristic by $(-1)^k$. It shows that the topology of M forces critical points to exist, whatever function we choose.

Example A.9.9.

1. The height function on the sphere S^2 has a minimum and a maximum: $\chi(S^2) = 1 - 0 + 1 = 2$.
2. The height function on an upright torus has a minimum, two saddle points and a maximum (Figure 3): $\chi = 1 - 2 + 1 = 0$. In fact, every Morse function on the torus has at least four critical points, at least two of which are saddle points (this follows from the Morse inequalities of Section A.10).
3. For the Rayleigh quotient on S^{n-1} (Example A.8.4), the two critical points $\pm\mathbf{q}_i$ have index $n - i$, so $\sum_k (-1)^k c_k = 2\sum_{i=1}^{n} (-1)^{n-i}$, which is 2 for odd n and 0 for even n. This agrees with the Euler characteristic of the sphere S^{n-1}, which is 2 if $n-1$ is even and 0 if $n-1$ is odd. For $n = 3$ and $A = \mathrm{diag}(3,2,1)$: two maxima $\pm\mathbf{e}_1$, two saddle points $\pm\mathbf{e}_2$, two minima $\pm\mathbf{e}_3$, and $2 - 2 + 2 = 2$.

A.10 Differential Forms and Stokes' Theorem

Lecture 3 contains several theorems of the same shape: the fundamental theorem for line integrals, Green's theorem and the divergence theorem (Theorems 14.8, 14.14 and 14.12 of Lecture 3) all say that the integral of a derivative over a region equals an integral over its boundary. The derivatives involved, the gradient, the curl and the divergence, look different, and the curl was only defined in two and three dimensions. Differential forms unify all of this: they are the natural integrands of k-dimensional integrals in $\mathbb{R}^n$, they have one derivative d, and they satisfy one theorem of Stokes. They are built from alternating tensors, the linear algebra of oriented areas and volumes. We use the tensors of Lecture 1 (Section 11 of Lecture 1), and we write $\mathbf{e}_1, \ldots, \mathbf{e}_n$ for the standard basis of $\mathbb{R}^n$. As in Lecture 1, elements of a general vector space V are written in normal font.

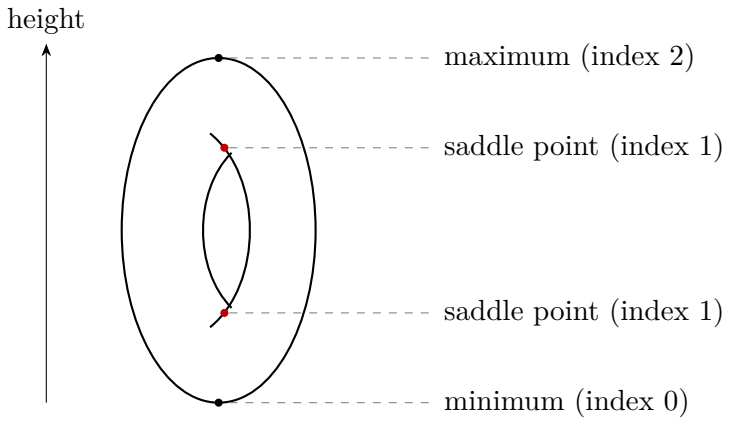


Figure 3: The height function on an upright torus has four critical points: the bottom (a minimum), the bottom and the top of the hole (two saddle points), and the top (a maximum). As the height c increases, the sublevel set is first a disk around the bottom (after the minimum), then a bent tube, i.e., a cylinder (after the first saddle point, where the disk closes up around the tube), then a torus with a hole at its top (after the second saddle point, where the two ends of the tube meet and close a loop around the hole of the torus), and finally the whole torus (after the maximum).

A.10.1 Alternating forms

Definition A.10.1 (Alternating form)**.** Let V be a finite-dimensional vector space and $k \geq 1$. A k-linear form ω on $V \times \cdots \times V$ (k factors) is alternating if $\omega(v_1, \ldots, v_k) = 0$ whenever $v_i = v_j$ for some $i \neq j$. The alternating k-forms form a subspace $\Lambda^k V^*$ of the space of all k-linear forms on V, which is $V^* \otimes \cdots \otimes V^*$ by Theorem 11.11 of Lecture 1. We also set $\Lambda^0 V^* = \mathbb{R}$. Note that $\Lambda^1 V^* = V^*$.

Lemma A.10.2. *Let ω be an alternating k-form on V.*

1. *Swapping two arguments changes the sign of $\omega(v_1, \ldots, v_k)$.*
2. *If $v_1, \ldots, v_k$ are linearly dependent, then $\omega(v_1, \ldots, v_k) = 0$. In particular, $\Lambda^k V^* = \{0\}$ for $k > \dim V$.*

Proof. (1) Put $v + w$ into both positions i and j, and expand by multilinearity:

$$0 = \omega(\ldots, v + w, \ldots, v + w, \ldots) = \omega(\ldots, v, \ldots, w, \ldots) + \omega(\ldots, w, \ldots, v, \ldots),$$

since the terms with v in both positions and with w in both positions vanish. (2) Some v_j is a linear combination of the others; by (1), we may assume that $j = k$, so $v_k = \sum_{i<k} a_i v_i$. Then $\omega(v_1, \ldots, v_k) = \sum_{i<k} a_i\, \omega(v_1, \ldots, v_{k-1}, v_i) = 0$, since each term contains the vector v_i twice. More than $\dim V$ vectors are always linearly dependent. □

The determinant of an $n \times n$ matrix, as a function of its columns, is an alternating n-form on $\mathbb{R}^n$: it is multilinear (Example 11.2 of Lecture 1), and it vanishes when two columns are equal, since swapping them changes its sign (Theorem 6.84 of Lecture 1). More generally, determinants produce alternating forms from covectors.

Definition A.10.3 (Wedge product of covectors)**.** For $\varphi_1, \ldots, \varphi_k \in V^*$, the wedge product $\varphi_1 \wedge \cdots \wedge \varphi_k \in \Lambda^k V^*$ is

$$(\varphi_1 \wedge \cdots \wedge \varphi_k)(v_1, \ldots, v_k) = \det\left(\varphi_i(v_j)\right)_{i,j=1,\ldots,k}.$$

This form is multilinear, since the j-th column of the matrix depends linearly on v_j and the determinant is linear in each column, and it is alternating, since two equal arguments give two equal columns. Swapping two covectors swaps two rows, so $\varphi \wedge \psi = -\psi \wedge \varphi$, and in particular $\varphi \wedge \varphi = 0$. For $k = 2$,

$$(\varphi \wedge \psi)(v, w) = \varphi(v)\psi(w) - \varphi(w)\psi(v), \qquad \text{i.e.,} \qquad \varphi \wedge \psi = \varphi \otimes \psi - \psi \otimes \varphi.$$

On $\mathbb{R}^n$, we write dx_i for the coordinate functional $\mathbf{x} \mapsto x_i$, i.e., for the dual basis vector e^i; the reason for this notation will appear soon. For an increasing sequence of indices $I = (i_1 < i_2 < \cdots < i_k)$, we write $dx_I = dx_{i_1} \wedge \cdots \wedge dx_{i_k}$. For example, $(dx_1 \wedge dx_2)(\mathbf{v}, \mathbf{w}) = v_1 w_2 - v_2 w_1$ is the signed area of the parallelogram spanned by the projections of $\mathbf{v}$ and $\mathbf{w}$ to the x_1x_2-plane, and $(dx_1 \wedge \cdots \wedge dx_n)(\mathbf{v}_1, \ldots, \mathbf{v}_n) = \det(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is the signed volume of the parallelepiped spanned by $\mathbf{v}_1, \ldots, \mathbf{v}_n$.

Theorem A.10.4. *The forms dx_I, for all increasing sequences $I = (i_1 < \cdots < i_k)$ in $\{1, \ldots, n\}$, form a basis of $\Lambda^k(\mathbb{R}^n)^*$. In particular, $\dim \Lambda^k(\mathbb{R}^n)^* = \binom{n}{k}$, and every alternating k-form on $\mathbb{R}^n$ is*

$$\omega = \sum_I \omega(\mathbf{e}_{i_1}, \ldots, \mathbf{e}_{i_k})\, dx_I.$$

Proof. For increasing sequences I and J, $dx_I(\mathbf{e}_{j_1}, \ldots, \mathbf{e}_{j_k}) = \det(\delta_{i_a j_b})_{a,b}$, which is 1 if $I = J$, and 0 otherwise: if $I \neq J$, some i_a is not among the j_b, and the a-th row of the matrix is zero. So applying a relation $\sum_I c_I\, dx_I = 0$ to $(\mathbf{e}_{j_1}, \ldots, \mathbf{e}_{j_k})$ gives $c_J = 0$, and the forms dx_I are linearly independent. Now let ω be alternating, and let $\eta = \sum_I \omega(\mathbf{e}_{i_1}, \ldots, \mathbf{e}_{i_k})\, dx_I$. The alternating forms ω and η agree on $(\mathbf{e}_{j_1}, \ldots, \mathbf{e}_{j_k})$ for every increasing J. By Lemma A.10.2, they also agree on all sequences of standard basis vectors with distinct indices, which are obtained from the increasing ones by swaps that change both values by the same sign, and both vanish on sequences with a repeated vector. A multilinear form is determined by its values on all sequences of basis vectors (Theorem 11.11 of Lecture 1), so $\omega = \eta$. □

For $k = n$, the theorem says that every alternating n-form on $\mathbb{R}^n$ is a multiple of the determinant $dx_1 \wedge \cdots \wedge dx_n$. The wedge product extends to forms of all degrees: for $\omega = \sum_I a_I\, dx_I \in \Lambda^k(\mathbb{R}^n)^*$ and $\eta = \sum_J b_J\, dx_J \in \Lambda^l(\mathbb{R}^n)^*$, let

$$\omega \wedge \eta = \sum_{I,J} a_I b_J\, dx_{i_1} \wedge \cdots \wedge dx_{i_k} \wedge dx_{j_1} \wedge \cdots \wedge dx_{j_l},$$

where each term is a wedge product of $k + l$ covectors, which is 0 if I and J have a common index, and $\pm dx_{I \cup J}$ otherwise. The wedge product is bilinear and associative, and since swapping two covectors changes the sign, moving each of the l covectors of dx_J past the k covectors of dx_I gives

$$\omega \wedge \eta = (-1)^{kl}\, \eta \wedge \omega.$$

For a number $a \in \Lambda^0$, we set $a \wedge \omega = a\omega$. So $dx_1 \wedge dx_2 = -dx_2 \wedge dx_1$ and $dx_1 \wedge dx_1 = 0$, but a 2-form can have a nonzero wedge product with itself (Exercise 16).

A.10.2 Differential forms and the exterior derivative

Definition A.10.5 (Differential form)**.** Let $U \subseteq \mathbb{R}^n$ be open. A differential k-form on U is a map ω that assigns to every point $\mathbf{x} \in U$ an alternating k-form $\omega_{\mathbf{x}} \in \Lambda^k(\mathbb{R}^n)^*$, such that the coefficients in

$$\omega = \sum_I f_I\, dx_I, \qquad \text{i.e.,} \qquad \omega_{\mathbf{x}} = \sum_I f_I(\mathbf{x})\, dx_I,$$

are smooth functions $f_I : U \to \mathbb{R}$. A 0-form is a smooth function. Forms are added, multiplied by functions and wedged pointwise.

Example A.10.6. 1. A 1-form $\omega = F_1\,dx_1 + \cdots + F_n\,dx_n$ satisfies $\omega_{\mathbf{x}}(\mathbf{v}) = F(\mathbf{x})^\top \mathbf{v}$, the work done by the vector field F along a small displacement $\mathbf{v}$ at $\mathbf{x}$. So 1-forms correspond to vector fields (Definition 13.1 of Lecture 3).

2. In $\mathbb{R}^3$, the 2-form $\omega = G_1\,dx_2 \wedge dx_3 + G_2\,dx_3 \wedge dx_1 + G_3\,dx_1 \wedge dx_2$ satisfies $\omega_{\mathbf{x}}(\mathbf{v}, \mathbf{w}) = G(\mathbf{x})^\top(\mathbf{v} \times \mathbf{w})$, since $(dx_2 \wedge dx_3)(\mathbf{v}, \mathbf{w}) = v_2 w_3 - v_3 w_2$ is the first entry of $\mathbf{v} \times \mathbf{w}$, and similarly for the others. This is the flux of G through the small parallelogram spanned by $\mathbf{v}$ and $\mathbf{w}$, so in $\mathbb{R}^3$, 2-forms also correspond to vector fields.

3. An n-form $f\,dx_1 \wedge \cdots \wedge dx_n$ is a function times the volume form $dx_1 \wedge \cdots \wedge dx_n$, like a density.

Definition A.10.7 (Exterior derivative)**.** The exterior derivative of a 0-form f is the 1-form

$$df = \frac{\partial f}{\partial x_1}\,dx_1 + \cdots + \frac{\partial f}{\partial x_n}\,dx_n,$$

so that $df_{\mathbf{x}}(\mathbf{v}) = Df(\mathbf{x})\mathbf{v}$ is the derivative of f in the direction $\mathbf{v}$. The exterior derivative of a k-form $\omega = \sum_I f_I\,dx_I$ is the $(k+1)$-form $d\omega = \sum_I df_I \wedge dx_I$.

For the coordinate function $f(\mathbf{x}) = x_i$, we get $df = dx_i$, which explains the notation. Since $dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ is $\pm dx_I$ or 0 for any (not necessarily increasing) indices, the formula $d(f\,dx_{i_1} \wedge \cdots \wedge dx_{i_k}) = df \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ holds for all indices, by linearity.

Proposition A.10.8. *1. d is linear.*

2. (Product rule.) $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k\,\omega \wedge d\eta$ for a k-form ω and any form η.

3. $d(d\omega) = 0$ for every form ω.

Proof. (1) is clear from the definition. (2) By linearity, it suffices to consider $\omega = f\,dx_I$ and $\eta = g\,dx_J$. Then $\omega \wedge \eta = fg\,dx_I \wedge dx_J$, and the product rule $d(fg) = g\,df + f\,dg$ gives

$$d(\omega \wedge \eta) = g\,df \wedge dx_I \wedge dx_J + f\,dg \wedge dx_I \wedge dx_J = d\omega \wedge \eta + (-1)^k\,\omega \wedge d\eta,$$

since moving the 1-form dg past the k covectors of dx_I gives the sign $(-1)^k$. (3) For $\omega = f\,dx_I$,

$$d(d\omega) = d\Big(\sum_j \frac{\partial f}{\partial x_j}\,dx_j \wedge dx_I\Big) = \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j}\,dx_i \wedge dx_j \wedge dx_I = 0,$$

since the second derivatives are symmetric in i and j (Theorem 8.2 of Lecture 3), while $dx_i \wedge dx_j = -dx_j \wedge dx_i$: the terms for (i,j) and (j,i) cancel, and the terms with $i = j$ vanish. □

Example A.10.9 (Vector calculus)**.** In $\mathbb{R}^3$, let $\omega_F = F_1\,dx_1 + F_2\,dx_2 + F_3\,dx_3$ and $\eta_G = G_1\,dx_2 \wedge dx_3 + G_2\,dx_3 \wedge dx_1 + G_3\,dx_1 \wedge dx_2$ be the 1-form and the 2-form of vector fields F and G, as in Example A.10.6. A short computation gives

$$\begin{aligned}
df &= \tfrac{\partial f}{\partial x_1}\,dx_1 + \tfrac{\partial f}{\partial x_2}\,dx_2 + \tfrac{\partial f}{\partial x_3}\,dx_3,\\
d\omega_F &= \Big(\tfrac{\partial F_3}{\partial x_2} - \tfrac{\partial F_2}{\partial x_3}\Big)dx_2 \wedge dx_3 + \Big(\tfrac{\partial F_1}{\partial x_3} - \tfrac{\partial F_3}{\partial x_1}\Big)dx_3 \wedge dx_1\\
&\quad + \Big(\tfrac{\partial F_2}{\partial x_1} - \tfrac{\partial F_1}{\partial x_2}\Big)dx_1 \wedge dx_2,\\
d\eta_G &= \Big(\tfrac{\partial G_1}{\partial x_1} + \tfrac{\partial G_2}{\partial x_2} + \tfrac{\partial G_3}{\partial x_3}\Big)\,dx_1 \wedge dx_2 \wedge dx_3.
\end{aligned}$$

With the correspondences of Example A.10.6, d is the gradient on 0-forms, the curl on 1-forms, and the divergence on 2-forms, and $d \circ d = 0$ contains the identities $\operatorname{curl} \nabla f = 0$ and $\operatorname{div} \operatorname{curl} F = 0$ (Proposition 13.8 of Lecture 3). In the plane, $d(F_1\,dx_1 + F_2\,dx_2) = \operatorname{curl} F\, dx_1 \wedge dx_2$, and in $\mathbb{R}^n$, $d(\sum_j F_j\,dx_j) = \sum_{i<j} (\frac{\partial F_j}{\partial x_i} - \frac{\partial F_i}{\partial x_j})\, dx_i \wedge dx_j$, whose coefficients are those of the skew-symmetric part of the Jacobian of F: this is the curl in every dimension.

A.10.3 Pullback and integration

Definition A.10.10 (Pullback)**.** Let $U \subseteq \mathbb{R}^m$ and $U' \subseteq \mathbb{R}^n$ be open, let $\boldsymbol{\phi} : U \to U'$ be smooth, and let ω be a k-form on U'. The pullback of ω is the k-form $\boldsymbol{\phi}^*\omega$ on U with

$$(\boldsymbol{\phi}^*\omega)_{\mathbf{x}}(\mathbf{v}_1, \ldots, \mathbf{v}_k) = \omega_{\boldsymbol{\phi}(\mathbf{x})}\big(D\boldsymbol{\phi}(\mathbf{x})\mathbf{v}_1, \ldots, D\boldsymbol{\phi}(\mathbf{x})\mathbf{v}_k\big),$$

and $\boldsymbol{\phi}^* f = f \circ \boldsymbol{\phi}$ for a 0-form f.

In words: to evaluate $\boldsymbol{\phi}^*\omega$ on vectors at $\mathbf{x}$, push the vectors forward with the derivative of $\boldsymbol{\phi}$, and evaluate ω on them at $\boldsymbol{\phi}(\mathbf{x})$. Writing $y_1, \ldots, y_n$ for the coordinates on U', we get

$$\boldsymbol{\phi}^*(dy_i) = d\phi_i = \sum_{j=1}^{m} \frac{\partial \phi_i}{\partial x_j}\, dx_j,$$

since $dy_i\big(D\boldsymbol{\phi}(\mathbf{x})\mathbf{v}\big) = D\phi_i(\mathbf{x})\mathbf{v}$. One checks that the pullback respects the wedge product, $\boldsymbol{\phi}^*(\omega \wedge \eta) = \boldsymbol{\phi}^*\omega \wedge \boldsymbol{\phi}^*\eta$, that $(\boldsymbol{\psi} \circ \boldsymbol{\phi})^* = \boldsymbol{\phi}^* \circ \boldsymbol{\psi}^*$ by the chain rule, and that the pullback commutes with the exterior derivative,

$$\boldsymbol{\phi}^*(d\omega) = d(\boldsymbol{\phi}^*\omega).$$

For 0-forms, this is the chain rule, $d(f \circ \boldsymbol{\phi}) = \boldsymbol{\phi}^*(df)$; for $\omega = f\,dy_I$, it follows with Proposition A.10.8, since $\boldsymbol{\phi}^*\omega = (f \circ \boldsymbol{\phi})\, d\phi_{i_1} \wedge \cdots \wedge d\phi_{i_k}$ and $d(d\phi_i) = 0$. So pullbacks are computed by substituting $y_i = \phi_i(\mathbf{x})$ and $dy_i = d\phi_i$. For forms of top degree, the Jacobian determinant appears.

Proposition A.10.11. *If $m = n$, then $\boldsymbol{\phi}^*(f\,dy_1 \wedge \cdots \wedge dy_n) = (f \circ \boldsymbol{\phi}) \det D\boldsymbol{\phi}\; dx_1 \wedge \cdots \wedge dx_n$.*

Proof. At each point, $\boldsymbol{\phi}^*(dy_1 \wedge \cdots \wedge dy_n)$ is an alternating n-form, so it is a multiple of $dx_1 \wedge \cdots \wedge dx_n$ (Theorem A.10.4), and the factor is its value on $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$:

$$(dy_1 \wedge \cdots \wedge dy_n)\big(D\boldsymbol{\phi}(\mathbf{x})\mathbf{e}_1, \ldots, D\boldsymbol{\phi}(\mathbf{x})\mathbf{e}_n\big) = \det\Big(\frac{\partial \phi_i}{\partial x_j}(\mathbf{x})\Big)_{i,j} = \det D\boldsymbol{\phi}(\mathbf{x}),$$

since $dy_i\big(D\boldsymbol{\phi}(\mathbf{x})\mathbf{e}_j\big)$ is the (i,j) entry of $D\boldsymbol{\phi}(\mathbf{x})$. □

For example, for polar coordinates $\boldsymbol{\phi}(r, \theta) = (r\cos\theta, r\sin\theta)$, we get $\boldsymbol{\phi}^*(dx_1 \wedge dx_2) = r\,dr \wedge d\theta$, the factor r of Example 4.7 of Lecture 3.

Definition A.10.12 (Integral of a differential form)**.**

1. Let $\omega = f\,dx_1 \wedge \cdots \wedge dx_k$ be a k-form on an open set that contains a box $Q \subseteq \mathbb{R}^k$. Then $\int_Q \omega = \int_Q f(\mathbf{x})\,d\mathbf{x}$, the multiple integral of Lecture 3.

2. A k-cube in an open set $U \subseteq \mathbb{R}^n$ is a smooth map $\mathbf{c} : [0,1]^k \to U$ (more precisely, the restriction of a smooth map on an open set that contains $[0,1]^k$). For a k-form ω on U, we define $\int_{\mathbf{c}} \omega = \int_{[0,1]^k} \mathbf{c}^*\omega$. A 0-cube is a point $\mathbf{c} \in U$, and $\int_{\mathbf{c}} f = f(\mathbf{c})$.

A k-cube is a parametrized k-dimensional piece of U: a curve for $k = 1$, a surface patch for $k = 2$. Integrals of k-forms do not depend on the parametrization, except for the orientation.

Proposition A.10.13. *Let $\boldsymbol{\psi}$ be a smooth bijection of $[0,1]^k$ onto itself with a smooth inverse. If $\det D\boldsymbol{\psi} > 0$ everywhere, then $\int_{\mathbf{c}\circ\boldsymbol{\psi}} \omega = \int_{\mathbf{c}} \omega$. If $\det D\boldsymbol{\psi} < 0$ everywhere, then $\int_{\mathbf{c}\circ\boldsymbol{\psi}} \omega = -\int_{\mathbf{c}} \omega$.*

Proof. Write $\mathbf{c}^*\omega = g\, dx_1 \wedge \cdots \wedge dx_k$. By Proposition A.10.11, $(\mathbf{c}\circ\boldsymbol{\psi})^*\omega = \boldsymbol{\psi}^*(\mathbf{c}^*\omega) = (g\circ\boldsymbol{\psi})\det D\boldsymbol{\psi}\; dx_1 \wedge \cdots \wedge dx_k$. Now the claim is the change of variables formula (Theorem 14.4 of Lecture 3), applied to the interior of the cube, in which $|\det D\boldsymbol{\psi}| = \pm \det D\boldsymbol{\psi}$. □

Example A.10.14. 1. (Line integrals.) For a curve $\mathbf{c} : [0,1] \to U$ and the 1-form $\omega = \sum_i F_i\, dx_i$, we get $\mathbf{c}^*\omega = \sum_i F_i(\mathbf{c}(t))\, c_i'(t)\, dt$, so

$$\int_{\mathbf{c}} \omega = \int_0^1 F(\mathbf{c}(t))^\top \mathbf{c}'(t)\, dt,$$

the line integral of the vector field F along $\mathbf{c}$ (Definition 14.7 of Lecture 3).

2. (Flux integrals.) For a surface patch $\mathbf{c} : [0,1]^2 \to \mathbb{R}^3$ and the 2-form ω of a vector field G as in Example A.10.6, the pullback is $\mathbf{c}^*\omega = G(\mathbf{c})^\top(\partial_1\mathbf{c} \times \partial_2\mathbf{c})\, du_1 \wedge du_2$, so $\int_{\mathbf{c}} \omega$ is the flux of G through the surface.

A.10.4 Manifolds with boundary and orientations

Stokes' theorem relates an integral over a k-dimensional region to an integral over its boundary, as for a ball and its boundary sphere, or for a surface and the curve around it. Such regions are manifolds with boundary: near their boundary points, they look like a half-space. Let

$$\mathbb{H}^k = \{\mathbf{u} \in \mathbb{R}^k : u_k \geq 0\}, \qquad \partial\mathbb{H}^k = \{\mathbf{u} \in \mathbb{R}^k : u_k = 0\}.$$

A set $V \subseteq \mathbb{H}^k$ is open in $\mathbb{H}^k$ if $V = V' \cap \mathbb{H}^k$ for an open set $V' \subseteq \mathbb{R}^k$, and a map on V is smooth if it is the restriction of a smooth map on such a set V'.

Definition A.10.15 (Submanifold with boundary)**.** Let $1 \leq k \leq n$. A set $M \subseteq \mathbb{R}^n$ is a k-dimensional submanifold with boundary if every $\mathbf{p} \in M$ has a local parametrization: a set V that is open in $\mathbb{R}^k$ or in $\mathbb{H}^k$, an open set $W \ni \mathbf{p}$, and a smooth map $\boldsymbol{\phi} : V \to \mathbb{R}^n$ that maps V homeomorphically onto $M \cap W$, with $\operatorname{rank} D\boldsymbol{\phi}(\mathbf{u}) = k$ for all $\mathbf{u} \in V$. The boundary ∂M is the set of the points $\boldsymbol{\phi}(\mathbf{u})$ with $\mathbf{u} \in \partial\mathbb{H}^k$, for the local parametrizations $\boldsymbol{\phi}$ on open subsets of $\mathbb{H}^k$.

With the inverse function theorem (Theorem A.5.6), one shows that the transition map $\boldsymbol{\phi}_2^{-1}\circ\boldsymbol{\phi}_1$ between two local parametrizations is smooth, with invertible derivatives, and that whether a point lies in ∂M does not depend on the parametrization. The maps $\mathbf{u}' \mapsto \boldsymbol{\phi}(\mathbf{u}', 0)$ are local parametrizations of ∂M by open subsets of $\mathbb{R}^{k-1}$, so ∂M is a $(k-1)$-dimensional submanifold with empty boundary. By Proposition A.7.2, the submanifolds of Definition A.7.1 are submanifolds with empty boundary.

Example A.10.16. 1. Let $g : \mathbb{R}^n \to \mathbb{R}$ be smooth, with $\nabla g(\mathbf{x}) \neq \mathbf{0}$ whenever $g(\mathbf{x}) = 0$. Then $M = \{\mathbf{x} : g(\mathbf{x}) \leq 0\}$ is an n-dimensional submanifold with boundary $\partial M = \{\mathbf{x} : g(\mathbf{x}) = 0\}$. Near a point $\mathbf{p}$ with $g(\mathbf{p}) < 0$, M contains an open set, on which the identity is a local parametrization. Near a point with $g(\mathbf{p}) = 0$, renumber the coordinates so that $\frac{\partial g}{\partial x_n}(\mathbf{p}) \neq 0$. Then $\boldsymbol{\Phi}(\mathbf{x}) = (x_1, \ldots, x_{n-1}, -g(\mathbf{x}))$ has an invertible derivative at $\mathbf{p}$, so it maps an open set $W \ni \mathbf{p}$ bijectively onto an open set V', with a smooth inverse. It maps $M \cap W$ onto $V' \cap \mathbb{H}^n$ and $\partial M \cap W$ onto $V' \cap \partial\mathbb{H}^n$, so the restriction of $\boldsymbol{\Phi}^{-1}$ to $V' \cap \mathbb{H}^n$ is a local parametrization. For $g(\mathbf{x}) = \|\mathbf{x}\|^2 - 1$, M is the closed unit ball B, with $\partial B = S^{n-1}$.

2. The closed upper hemisphere $S^2_+ = \{\mathbf{x} \in S^2 : x_3 \geq 0\}$ is a 2-dimensional submanifold with boundary, whose boundary is the equator $\{\mathbf{x} \in S^2 : x_3 = 0\}$. Near the equator, the longitude θ (in an open interval of length less than 2π) and the latitude $s \in [0, \frac{\pi}{2})$ give the local parametrization $\boldsymbol{\phi}(\theta, s) = (\cos s \cos\theta,\, \cos s \sin\theta,\, \sin s)$.

3. A compact interval $[a, b]$ is a 1-dimensional submanifold of $\mathbb{R}$ with boundary $\{a, b\}$, and the sphere S^{n-1} and the orthogonal group $\mathrm{O}(n)$ have empty boundary. The square $[0, 1]^2$ is not a submanifold with boundary, because of its corners; Stokes' theorem for cubes below covers it.

By Proposition A.10.13, integrals of k-forms change their sign under reparametrizations with negative Jacobian determinants. So to integrate over M, we must choose its parametrizations consistently.

Definition A.10.17 (Orientation)**.** An orientation of a k-dimensional submanifold M with boundary is a collection of local parametrizations whose images cover M, such that the transition map between any two of them has a positive Jacobian determinant everywhere. These parametrizations, and all others whose transition maps with them have positive Jacobian determinants, are called positively oriented; those whose transition maps with them have negative Jacobian determinants are negatively oriented. M is orientable if it has an orientation, and oriented if one has been chosen.

Example A.10.18. 1. An n-dimensional submanifold $M \subseteq \mathbb{R}^n$ with boundary has the standard orientation, for which the positively oriented parametrizations are those with $\det D\boldsymbol{\phi} > 0$.

2. A 2-dimensional submanifold $S \subseteq \mathbb{R}^3$ with boundary (a surface) is oriented by a continuous unit normal field $\mathbf{n}$: the positively oriented parametrizations are those for which $\partial_1\boldsymbol{\phi} \times \partial_2\boldsymbol{\phi}$ is a positive multiple of $\mathbf{n}$. Conversely, every orientation of S comes from such a normal field. For example, S^2 and S^2_+ are oriented by the outward normal $\mathbf{n}(\mathbf{x}) = \mathbf{x}$, and the parametrization $\boldsymbol{\phi}(\theta, s)$ of Example A.10.16 is positively oriented, since $\partial_\theta\boldsymbol{\phi} \times \partial_s\boldsymbol{\phi} = \cos s\ \boldsymbol{\phi}(\theta, s)$. The Möbius strip is not orientable: a normal vector that is moved continuously once around the strip comes back as its negative.

Definition A.10.19 (Induced orientation)**.** Let M be an oriented k-dimensional submanifold with boundary, where $k \geq 2$. The induced orientation of ∂M is the one for which, for every positively oriented local parametrization $\boldsymbol{\phi}$ of M on an open subset of $\mathbb{H}^k$, the parametrization $\boldsymbol{\phi}_\partial(\mathbf{u}') = \boldsymbol{\phi}(\mathbf{u}', 0)$ of ∂M is positively oriented if k is even, and negatively oriented if k is odd. For $k = 1$ and a compact M, ∂M is a finite set of points, and the induced orientation gives a point $\boldsymbol{\phi}(0)$ the sign -1 if $\boldsymbol{\phi}$ is positively oriented, and $+1$ if it is negatively oriented; the integral of a function f over ∂M is the sum of the values $\pm f(\mathbf{p})$ over $\mathbf{p} \in \partial M$, with these signs.

(This is consistent: a transition map $\boldsymbol{\psi}$ between two parametrizations on open subsets of $\mathbb{H}^k$ maps points of $\mathbb{H}^k$ to $\mathbb{H}^k$ and points of $\partial\mathbb{H}^k$ to $\partial\mathbb{H}^k$, so at a point of $\partial\mathbb{H}^k$, the last row of $D\boldsymbol{\psi}$ is $(0, \ldots, 0, \frac{\partial\psi_k}{\partial u_k})$ with $\frac{\partial\psi_k}{\partial u_k} > 0$, and the restriction of $\boldsymbol{\psi}$ to $\partial\mathbb{H}^k$ also has a positive Jacobian determinant.) In words, a basis $\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}$ of the boundary is positively oriented if $\mathbf{n}, \mathbf{v}_1, \ldots, \mathbf{v}_{k-1}$ is positively oriented for an outward vector $\mathbf{n}$: "outward normal first". For $\mathbb{H}^k$, the vector $-\mathbf{e}_k$ points outward, and $\det(-\mathbf{e}_k, \mathbf{e}_1, \ldots, \mathbf{e}_{k-1}) = (-1)^k$. For a region $M \subseteq \mathbb{R}^2$ with the standard orientation, ∂M is traversed with M on its left, i.e., counterclockwise for a disk, as in Green's theorem. For a region $M \subseteq \mathbb{R}^3$, the positively oriented parametrizations $\boldsymbol{\psi}$ of ∂M are those for which $\partial_1\boldsymbol{\psi} \times \partial_2\boldsymbol{\psi}$ points outward, as in the divergence theorem. For a surface S oriented by a normal field $\mathbf{n}$, ∂S is traversed with S on the left of a walker whose head points in the direction of $\mathbf{n}$. For example, the equator of S^2_+ with the outward normal is traversed counterclockwise, as seen from above: $\boldsymbol{\phi}_\partial(\theta) = (\cos\theta, \sin\theta, 0)$.

Definition A.10.20 (Integrals over manifolds)**.** Let M be a compact oriented k-dimensional submanifold with boundary, and let ω be a k-form on an open set that contains M.

1. Suppose that ω vanishes at all points of M outside $\boldsymbol{\phi}(K)$, for a positively oriented local parametrization $\boldsymbol{\phi} : V \to M \cap W$ and a compact set $K \subseteq V$. Then $\boldsymbol{\phi}^*\omega = g\, du_1 \wedge \cdots \wedge du_k$, where g vanishes on V outside K, and we define $\int_M \omega = \int_V g(\mathbf{u})\, d\mathbf{u}$, the integral of g, extended by 0, over a box that contains K.

2. In general, let $\rho_1, \ldots, \rho_m \geq 0$ be smooth functions on $\mathbb{R}^n$ with $\sum_i \rho_i = 1$ on an open set that contains M, such that each ρ_i vanishes at all points of M outside $\boldsymbol{\phi}_i(K_i)$, for some $\boldsymbol{\phi}_i$ and K_i as in 1 (a partition of unity). We define $\int_M \omega = \sum_{i=1}^m \int_M \rho_i \omega$.

In 1, the integral does not depend on $\boldsymbol{\phi}$: for a second positively oriented parametrization, the two integrals agree by Proposition A.10.11 and the change of variables formula, applied to the transition map, as in the proof of Proposition A.10.13. For a negatively oriented $\boldsymbol{\phi}$, the same argument gives $\int_M \omega = -\int_V \boldsymbol{\phi}^*\omega$. Partitions of unity exist because M is compact: M is covered by finitely many open balls whose closures lie in sets W with positively oriented parametrizations, and dividing smooth bump functions that are positive exactly on these balls (such as $e^{-1/(1-\|\mathbf{x}\|^2)}$ on the unit ball) by their sum gives the functions ρ_i; we omit the details. The definition in 2 does not depend on the partition of unity: for a second one, $\sigma_1, \ldots, \sigma_l$, both sums are equal to $\sum_{i,j} \int_M \rho_i \sigma_j \omega$. Integrals over ∂M are defined in the same way, with the induced orientation.

For $k = n$ and the standard orientation, Proposition A.10.11 and the change of variables formula give $\int_M f\, dx_1 \wedge \cdots \wedge dx_n = \int_M f(\mathbf{x})\, d\mathbf{x}$, the multiple integral of Lecture 3. For a surface $S \subseteq \mathbb{R}^3$ oriented by $\mathbf{n}$ and the 2-form ω of a vector field G, $\int_S \omega$ is the flux of G through S in the direction of $\mathbf{n}$, by Example A.10.14.

A.10.5 Stokes' theorem

Theorem A.10.21 (Stokes)**.** *Let $M \subseteq \mathbb{R}^n$ be a compact oriented k-dimensional submanifold with boundary, where $k \geq 1$, let ∂M have the induced orientation, and let ω be a $(k-1)$-form on an open set that contains M. Then*

$$\int_M d\omega = \int_{\partial M} \omega.$$

In particular, $\int_M d\omega = 0$ if ∂M is empty.

We prove the theorem for cubes, which contains all the analysis, and then explain how the general case reduces to cubes. The boundary of the square $[0,1]^2$ consists of four edges, which we traverse counterclockwise. In general, the boundary of a k-cube $\mathbf{c}$ consists of its $2k$ faces, with signs. For $j = 1, \ldots, k$ and $\alpha \in \{0, 1\}$, the face $\mathbf{c}_{(j,\alpha)}$ is the $(k-1)$-cube obtained by fixing the j-th coordinate to α:

$$\mathbf{c}_{(j,\alpha)}(u_1, \ldots, u_{k-1}) = \mathbf{c}(u_1, \ldots, u_{j-1}, \alpha, u_j, \ldots, u_{k-1}).$$

The boundary of $\mathbf{c}$ is the formal sum $\partial\mathbf{c} = \sum_{j=1}^k \sum_{\alpha=0,1} (-1)^{j+\alpha}\, \mathbf{c}_{(j,\alpha)}$, and for a $(k-1)$-form ω, we define

$$\int_{\partial \mathbf{c}} \omega = \sum_{j=1}^{k} \sum_{\alpha=0,1} (-1)^{j+\alpha} \int_{\mathbf{c}_{(j,\alpha)}} \omega.$$

For a curve ($k = 1$), $\int_{\partial \mathbf{c}} f = f(\mathbf{c}(1)) - f(\mathbf{c}(0))$: the boundary is the end point minus the starting point. For the square, the edges $(u, 0)$, $(1, u)$, $(u, 1)$ and $(0, u)$ get the signs $+$, $+$, $-$ and $-$, so they are traversed counterclockwise, as in Green's theorem.

Theorem A.10.22 (Stokes' theorem for cubes)**.** *Let ω be a $(k-1)$-form on an open set $U \subseteq \mathbb{R}^n$, and let $\mathbf{c}$ be a k-cube in U. Then*

$$\int_{\mathbf{c}} d\omega = \int_{\partial \mathbf{c}} \omega.$$

Proof. First, let $\mathbf{c}$ be the identity map of $[0,1]^k$, so that ω is a $(k-1)$-form on an open set containing $[0,1]^k \subseteq \mathbb{R}^k$. By linearity, we may assume that $\omega = f\, du_1 \wedge \cdots \wedge \widehat{du_j} \wedge \cdots \wedge du_k$, where the hat means that du_j is left out. Then

$$d\omega = \frac{\partial f}{\partial u_j}\, du_j \wedge du_1 \wedge \cdots \wedge \widehat{du_j} \wedge \cdots \wedge du_k = (-1)^{j-1} \frac{\partial f}{\partial u_j}\, du_1 \wedge \cdots \wedge du_k,$$

since the other terms of df contain some du_i twice, and moving du_j to its place takes $j-1$ swaps. By Fubini's theorem and the fundamental theorem of calculus, integrating over u_j first,

$$\int_{[0,1]^k} d\omega = (-1)^{j-1} \int_{[0,1]^{k-1}} \Big(f(\dots, \underset{j}{1}, \dots) - f(\dots, \underset{j}{0}, \dots) \Big)\, d\mathbf{u}',$$

where the j-th argument is 1 or 0, and $\mathbf{u}'$ denotes the other $k-1$ variables. On the other side, the i-th coordinate is constant on the face $\mathbf{c}_{(i,\alpha)}$, so the pullback of du_i to this face is 0. Hence only the two faces with $i = j$ contribute, and the pullback of ω to them is $f(\dots, \alpha, \dots)\, du_1 \wedge \cdots \wedge du_{k-1}$. So

$$\int_{\partial \mathbf{c}} \omega = (-1)^{j+1} \int_{[0,1]^{k-1}} f(\dots, \underset{j}{1}, \dots)\, d\mathbf{u}' + (-1)^{j} \int_{[0,1]^{k-1}} f(\dots, \underset{j}{0}, \dots)\, d\mathbf{u}',$$

which is the same. For a general k-cube $\mathbf{c}$, the faces of $\mathbf{c}$ are $\mathbf{c} \circ \mathbf{i}_{(j,\alpha)}$, where the $\mathbf{i}_{(j,\alpha)}$ are the faces of the identity cube $\mathbf{i}$ of $[0,1]^k$. Since the pullback commutes with d, the first case gives

$$\int_{\mathbf{c}} d\omega = \int_{[0,1]^k} \mathbf{c}^*(d\omega) = \int_{[0,1]^k} d(\mathbf{c}^*\omega) = \int_{\partial \mathbf{i}} \mathbf{c}^*\omega = \int_{\partial \mathbf{c}} \omega. \qquad \square$$

A k-cube need not be injective, so Theorem A.10.22 is not a special case of Theorem A.10.21; conversely, $[0,1]^k$ is not a submanifold with boundary for $k \geq 2$. But the analysis is the same, and Theorem A.10.21 reduces to Theorem A.10.22.

Idea of the proof of Theorem A.10.21. For $k = 1$, M consists of finitely many closed curves and arcs. Each of them can be parametrized, in the direction of its orientation, by a 1-cube $\mathbf{c}$ that is injective, except that $\mathbf{c}(0) = \mathbf{c}(1)$ for a closed curve, and Theorem A.10.22 gives $\int_{\mathbf{c}} df = f(\mathbf{c}(1)) - f(\mathbf{c}(0))$, the signed sum of f over the end points. Now let $k \geq 2$.

1. (Localization.) For a partition of unity $\rho_1, \dots, \rho_m$ as in Definition A.10.20, we have $\omega = \sum_i \rho_i \omega$ and $d\omega = \sum_i d(\rho_i \omega)$ near M, and both sides of Stokes' formula are linear in ω. So we may assume that ω and $d\omega$ vanish at all points of M outside $\boldsymbol{\phi}(K)$, for a positively oriented local parametrization $\boldsymbol{\phi} : V \to M \cap W$ and a compact set $K \subseteq V$. Shrinking and translating V if necessary, we may assume that V is open in $\mathbb{H}^k$.

2. (Pullback.) Since the pullback commutes with d, $\int_M d\omega = \int_V \boldsymbol{\phi}^*(d\omega) = \int_V d\eta$ for the $(k-1)$-form $\eta = \boldsymbol{\phi}^*\omega$, which vanishes outside K. On the boundary, the pullback of η to $\partial \mathbb{H}^k$ is $\boldsymbol{\phi}^*_\partial \omega$, so $\int_{\partial M} \omega$ is also an integral of η.

3. (A box.) Choose $a > 0$ such that K lies in the box $Q = [-a, a]^{k-1} \times [0, a]$, away from all its faces except the bottom face $u_k = 0$. The box Q is the image of the k-cube $\mathbf{A}(\mathbf{t}) =$

$(2at_1 - a, \ldots, 2at_{k-1} - a,\ at_k)$, an affine map with a positive Jacobian determinant. Extending η by 0, Proposition A.10.11, the change of variables formula and Theorem A.10.22 give

$$\int_M d\omega = \int_Q d\eta = \int_{\mathbf{A}} d\eta = \int_{\partial \mathbf{A}} \eta = (-1)^{k+0} \int_{\mathbf{A}_{(k,0)}} \eta,$$

since η vanishes on the other $2k-1$ faces of $\mathbf{A}$. The bottom face $\mathbf{A}_{(k,0)}(\mathbf{t}') = (2at_1 - a, \ldots, 2at_{k-1} - a,\ 0)$ maps $[0,1]^{k-1}$ onto $[-a,a]^{k-1} \times \{0\}$ with a positive Jacobian determinant, so the right side is $(-1)^k \int_{[-a,a]^{k-1}} \boldsymbol{\phi}_\partial^* \omega$. This is $\int_{\partial M} \omega$, since $\boldsymbol{\phi}_\partial$ is positively oriented if k is even, and negatively oriented if k is odd (Definition A.10.19). If K does not meet $\partial \mathbb{H}^k$, both sides are 0.

This is where the sign $(-1)^k$ in the induced orientation comes from: it is the sign of the bottom face of a cube. Instead of a partition of unity, one can also cut M into pieces parametrized by cubes, as for the divergence theorem in Lecture 3. The integrals over the faces shared by two neighboring pieces cancel, since the two pieces induce opposite orientations on them, and only the integral over ∂M remains. □

All the integral theorems of Lecture 3 are special cases of Theorem A.10.21.

- For $k = 1$ and a 0-form f: if M is an arc from $\mathbf{p}$ to $\mathbf{q}$, then $\int_M df = f(\mathbf{q}) - f(\mathbf{p})$, the fundamental theorem for line integrals. For $M = [a,b] \subseteq \mathbb{R}$, it is the fundamental theorem of calculus.

- For a compact region $M \subseteq \mathbb{R}^2$ with smooth boundary (a compact 2-dimensional submanifold with boundary, such as the sets $\{g \leq 0\}$ of Example A.10.16), with the standard orientation, and for $\omega = F_1\, dx_1 + F_2\, dx_2$: $d\omega = \operatorname{curl} F\, dx_1 \wedge dx_2$, and Stokes' theorem is Green's theorem, with ∂M traversed counterclockwise, i.e., with M on its left.

- For a compact region $M \subseteq \mathbb{R}^3$ with smooth boundary and the 2-form ω of a vector field G: $d\omega = \operatorname{div} G\, dx_1 \wedge dx_2 \wedge dx_3$, the integral $\int_{\partial M} \omega$ is the flux of G through ∂M with the outward normal, and Stokes' theorem is the divergence theorem.

- For a compact surface $S \subseteq \mathbb{R}^3$ with boundary, oriented by a unit normal field $\mathbf{n}$, and for $\omega = F_1\, dx_1 + F_2\, dx_2 + F_3\, dx_3$: $d\omega$ is the 2-form of the vector field $\operatorname{curl} F$, and Stokes' theorem is the classical Stokes theorem: the circulation of F around ∂S equals the flux of $\operatorname{curl} F$ through S in the direction of $\mathbf{n}$.

Example A.10.23. 1. The 2-form $\omega = x_1\, dx_2 \wedge dx_3 + x_2\, dx_3 \wedge dx_1 + x_3\, dx_1 \wedge dx_2$ of the vector field $G(\mathbf{x}) = \mathbf{x}$ has $d\omega = 3\, dx_1 \wedge dx_2 \wedge dx_3$. For the closed unit ball $B \subseteq \mathbb{R}^3$, whose boundary S^2 gets the outward normal, Stokes' theorem gives $\int_{S^2} \omega = 3 \cdot \frac{4}{3}\pi = 4\pi$, as in Example 14.13 of Lecture 3.

2. Let S^2_+ be the closed upper hemisphere with the outward normal, and let $\omega = -x_2\, dx_1 + x_1\, dx_2$ be the 1-form of $F(\mathbf{x}) = (-x_2, x_1, 0)$. Then $d\omega = 2\, dx_1 \wedge dx_2$, the 2-form of $\operatorname{curl} F = (0,0,2)$. The equator is traversed counterclockwise by $\boldsymbol{\phi}_\partial(\theta) = (\cos\theta, \sin\theta, 0)$, with $\boldsymbol{\phi}_\partial^* \omega = (\sin^2\theta + \cos^2\theta)\, d\theta$, so $\int_{S^2_+} d\omega = \int_0^{2\pi} d\theta = 2\pi$. The closed unit disk D in the plane $x_3 = 0$, oriented by the upward normal $\mathbf{e}_3$, has the same oriented boundary, so $\int_D d\omega = 2\pi$ as well, which is twice the area of D. In general, the flux of $\operatorname{curl} F$ through an oriented surface only depends on its oriented boundary.

A.10.6 Closed and exact forms

Definition A.10.24. A k-form ω is closed if $d\omega = 0$, and exact if $\omega = d\eta$ for a $(k-1)$-form η.

Every exact form is closed, since $d(d\eta) = 0$. For a 1-form $\omega = \sum_i F_i\, dx_i$, exact means that $F = \nabla f$ is a gradient field, and closed means that $\frac{\partial F_j}{\partial x_i} = \frac{\partial F_i}{\partial x_j}$ for all i and j, i.e., that the Jacobian of F is symmetric. The Poincaré lemma of Lecture 3 (Theorem 14.10) says that on $\mathbb{R}^n$, every closed 1-form is exact. More generally, on a star-shaped open set, every closed k-form with $k \geq 1$ is exact. On other sets, this can fail, and the failure detects holes.

Example A.10.25 (The angle form)**.** On $U = \mathbb{R}^2 \setminus \{\mathbf{0}\}$, the 1-form

$$\omega = \frac{-x_2\, dx_1 + x_1\, dx_2}{x_1^2 + x_2^2}$$

is closed: it belongs to the vortex field of Example 14.11 of Lecture 3, whose curl is 0. It is not exact: for the circle $\mathbf{c}(t) = (\cos 2\pi t, \sin 2\pi t)$, the pullback is $\mathbf{c}^*\omega = 2\pi\, dt$, so $\int_{\mathbf{c}} \omega = 2\pi$, while $\int_{\mathbf{c}} df = f(\mathbf{c}(1)) - f(\mathbf{c}(0)) = 0$ for every function f, by Theorem A.10.22, since $\mathbf{c}(1) = \mathbf{c}(0)$. Near every point of U, $\omega = d\theta$ for the polar angle θ, which explains its name; but the angle cannot be defined as a smooth function on all of U. For every closed curve $\mathbf{c}$ in U, the number $\frac{1}{2\pi}\int_{\mathbf{c}} \omega$ is an integer, the winding number of $\mathbf{c}$ around the origin.

Example A.10.26 (The solid angle form)**.** On $U = \mathbb{R}^3 \setminus \{\mathbf{0}\}$, let $\tau = \omega/\|\mathbf{x}\|^3$, where ω is the 2-form of Example A.10.23. It is the 2-form of the vector field $G(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|^3$, the electric field of a point charge (up to a constant), and it is closed, since $\operatorname{div} G = \sum_i \left(\|\mathbf{x}\|^{-3} - 3x_i^2\|\mathbf{x}\|^{-5}\right) = 0$. For a small parallelogram spanned by $\mathbf{v}$ and $\mathbf{w}$ at $\mathbf{x}$, $\tau_{\mathbf{x}}(\mathbf{v}, \mathbf{w})$ is approximately the solid angle under which it is seen from the origin, hence the name. Since $\|\mathbf{x}\| = 1$ on S^2, the pullbacks of τ and ω to S^2 agree, so $\int_{S^2} \tau = 4\pi$. But if $\tau = d\eta$ for a 1-form η on U, then Theorem A.10.21 for $S^2 \subseteq U$, which has empty boundary, would give $\int_{S^2} \tau = 0$. So τ is closed but not exact: it detects the hole of U with a sphere, as the angle form detects the hole of $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ with a circle.

The closed k-forms on U form a vector space, the kernel of d, and the exact k-forms form a subspace of it, the image of d. The quotient space (Section 9 of Lecture 1)

$$H^k(U) = \{\text{closed } k\text{-forms on } U\}/\{\text{exact } k\text{-forms on } U\}$$

is the k-th de Rham cohomology space of U. It measures how far closed forms are from being exact:

- $H^0(U)$ consists of the functions with $df = 0$, i.e., of the functions that are constant on each connected component of U (Section A.4). So $\dim H^0(U)$ is the number of connected components of U.
- By the Poincaré lemma, $H^k(\mathbb{R}^n) = \{0\}$ for $k \geq 1$.
- $H^1(\mathbb{R}^2 \setminus \{\mathbf{0}\})$ is one-dimensional, spanned by the class of the angle form: every closed 1-form η on $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ is $c\,\omega + df$ for some function f, with $c = \frac{1}{2\pi}\int_{\mathbf{c}} \eta$ for the unit circle $\mathbf{c}$.
- $H^2(\mathbb{R}^3 \setminus \{\mathbf{0}\}) \neq \{0\}$, by Example A.10.26. In fact, it is one-dimensional, spanned by the class of the solid angle form, while $H^1(\mathbb{R}^3 \setminus \{\mathbf{0}\}) = \{0\}$: closed curves cannot detect a point removed from $\mathbb{R}^3$.

De Rham's theorem says that these spaces only depend on the topology of U (more precisely, on its homotopy type), although they are defined with derivatives. They can also be defined for manifolds. For a compact manifold M, their dimensions $b_k = \dim H^k(M)$, the Betti numbers, refine the Euler characteristic: $\chi(M) = \sum_k (-1)^k b_k$. Together with the Morse relation (Theorem A.9.8), this leads to the Morse inequalities $c_k \geq b_k$: a Morse function on M has at least b_k critical points of index k. For example, the torus has $b_0 = 1$, $b_1 = 2$ and $b_2 = 1$, so every Morse function on it has at least four critical points, at least two of which are saddle points.

Remark A.10.27 (Differential forms in machine learning). Differential forms appear in machine learning in several places.

1. Probability densities transform like n-forms. If $\mathbf{y} = \boldsymbol{\phi}(\mathbf{x})$ for a diffeomorphism $\boldsymbol{\phi}$ with $\det D\boldsymbol{\phi} > 0$, Proposition A.10.11 gives $\boldsymbol{\phi}^*(p_Y \, dy_1 \wedge \cdots \wedge dy_n) = (p_Y \circ \boldsymbol{\phi}) \det D\boldsymbol{\phi} \, dx_1 \wedge \cdots \wedge dx_n$, which is the change of variables formula $p_X = (p_Y \circ \boldsymbol{\phi}) \det D\boldsymbol{\phi}$ behind normalizing flows (Remark 4.8 of Lecture 3).

2. Hamiltonian Monte Carlo samples from a density proportional to $e^{-U(\mathbf{q})}$ by simulating the motion of a particle with position $\mathbf{q}$ and momentum $\mathbf{p}$ in the potential U. The flow of these equations preserves the 2-form $\sum_i dq_i \wedge dp_i$ (it is symplectic), and hence volume (compare Liouville's formula, Proposition 16.1 of Lecture 3). The leapfrog integrator used in practice is symplectic as well, which keeps the error in the energy small over long trajectories, and the acceptance rate of the method high.

3. On a graph, functions on the nodes, on the edges and on the triangles are discrete analogues of 0-, 1- and 2-forms, the incidence matrices play the role of d, and $d \circ d = 0$ still holds. The resulting decomposition of flows on the edges into a gradient part, a curl part and a harmonic part (Hodge theory) is used, for example, to rank items from pairwise comparisons, and in neural networks on simplicial complexes.

A.11 Topology in Machine Learning

We end with a few places where the ideas of this appendix appear in machine learning. Some of them are theorems, others are well-supported heuristics and active research topics.

- **The manifold hypothesis.** High-dimensional data, such as images with millions of pixels, are believed to lie close to submanifolds of much smaller dimension: the images of a face, for example, vary with a small number of parameters such as pose, lighting and expression. PCA finds the best approximating linear subspace (Lecture 2); autoencoders and other nonlinear methods try to find curved, manifold-like structures.

- **Normalizing flows and topology.** A normalizing flow maps a simple distribution, such as a Gaussian on $\mathbb{R}^n$, to a complicated one by a diffeomorphism (Corollary A.5.5 and Remark 4.8 of Lecture 3). Diffeomorphisms preserve topology. So, by Proposition A.4.2, a flow cannot map $\mathbb{R}^n$, which is connected, exactly onto a distribution supported on two separate pieces, such as two clusters: it can only approximate it by squeezing a little probability mass through thin bridges between them.

- **Symmetries make critical points degenerate.** If a loss is invariant under a continuous family of transformations of the parameters, its minima are not isolated. For example, the ReLU satisfies $\max(0, ct) = c \max(0, t)$ for $c > 0$, so multiplying the incoming weights of

a hidden unit by c and its outgoing weights by $\frac{1}{c}$ does not change the network. The set of minima then contains curves (in the simplest case, the hyperbola $w_1 w_2 = 1$ for the loss $(w_1 w_2 - 1)^2$), and by Proposition A.9.4, the Hessian at a minimum must have the eigenvalue 0, in the tangent directions of these curves. Loss functions of neural networks are therefore not Morse functions.

- **Saddle points in high dimensions.** At a critical point of a loss with millions of parameters, all eigenvalues of the Hessian must be nonnegative for a local minimum. A heuristic, supported by experiments and by random matrix theory, says that critical points with high loss usually have many negative eigenvalues, i.e., a large index: they are saddle points rather than bad local minima. Gradient descent with noise, as in SGD, tends to escape from saddle points, which is part of the explanation why training works.

- **Connected sublevel sets.** Experiments show that two minima found by training the same network twice can often be joined by a path along which the loss stays low ("mode connectivity"): the sublevel sets $\{L \le c\}$ for small c seem to be connected, up to the symmetries of the network. Morse theory gives a language for such statements: a sublevel set can only become connected by passing saddle points of index 1.

A.12 Banach Spaces

In Lectures 2 and 4, we met vector spaces whose elements are functions, such as polynomials, continuous functions and random variables. These spaces are infinite-dimensional, and Remark A.3.11 showed that some results of the previous sections fail in them. Functional analysis studies such spaces. Its starting point is the property that makes $\mathbb{R}^n$ so convenient for analysis: completeness (Corollary A.3.2).

Definition A.12.1 (Normed and Banach spaces)**.** A norm on a real vector space V is a function $\|\cdot\| : V \to [0, \infty)$ such that $\|v\| = 0$ only for $v = 0$, $\|cv\| = |c|\|v\|$ and $\|u + v\| \le \|u\| + \|v\|$ for all $u, v \in V$ and $c \in \mathbb{R}$. A normed space is a metric space with the metric $d(u, v) = \|u - v\|$. It is a Banach space if it is complete, i.e., if every Cauchy sequence in it converges.

By Corollary A.3.2 and Theorem A.3.10, every finite-dimensional normed space is a Banach space, since a basis identifies it with $\mathbb{R}^n$.

Example A.12.2 (Banach spaces)**.**

1. The space $C[a, b]$ of continuous functions $f : [a, b] \to \mathbb{R}$ with the maximum norm $\|f\|_\infty = \max_{a \le x \le b} |f(x)|$ is a Banach space (Proposition A.12.3 below). Convergence in this norm is uniform convergence: $\|f_k - f\|_\infty \to 0$ means that $f_k(x) \to f(x)$ at the same speed for all x.

2. For $1 \le p < \infty$, the sequences $x = (x_1, x_2, \ldots)$ with $\|x\|_p = (\sum_k |x_k|^p)^{1/p} < \infty$ form a Banach space ℓ^p, and the bounded sequences with $\|x\|_\infty = \sup_k |x_k|$ form a Banach space ℓ^∞.

3. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $1 \le p < \infty$. The measurable functions with $\|f\|_p = (\int |f|^p \, d\mu)^{1/p} < \infty$, where functions that agree almost everywhere are identified, form a Banach space $L^p(\mu)$ (the Riesz–Fischer theorem, which we do not prove). The integral is the Lebesgue integral (Definition B.3.1 of the appendix to Lecture 4). For a probability measure, these are the spaces L^p of random variables of the appendix to Lecture 4; for the Lebesgue measure on $[a, b]$, we write $L^p[a, b]$; and for the counting measure on $\{1, 2, \ldots\}$, $L^p(\mu)$ is ℓ^p.

4. Not every normed space is complete. On $C[0,1]$, $\|f\|_1 = \int_0^1 |f(x)|\,dx$ is also a norm. The ramps f_k that are 0 on $[0, \frac{1}{2}]$, 1 on $[\frac{1}{2}+\frac{1}{k}, 1]$ and linear in between satisfy $\|f_k - f_l\|_1 \le \frac{1}{2\min(k,l)}$, so they form a Cauchy sequence for this norm. But they converge in this norm to the step function $\mathbf{1}_{(1/2,1]}$, and no continuous function is their limit. So $(C[0,1], \|\cdot\|_1)$ is not complete. Adding the limits of all Cauchy sequences, as the real numbers complete the rational numbers, gives $L^1[0,1]$.

Proposition A.12.3. *$C[a,b]$ with the maximum norm is complete.*

Proof. Let (f_k) be a Cauchy sequence. For each x, $|f_k(x) - f_l(x)| \le \|f_k - f_l\|_\infty$, so $(f_k(x))$ is a Cauchy sequence of real numbers, with a limit $f(x)$. Given $\varepsilon > 0$, choose N with $\|f_k - f_l\|_\infty \le \varepsilon$ for $k, l \ge N$. Letting $l \to \infty$ gives $|f_k(x) - f(x)| \le \varepsilon$ for all x and $k \ge N$, so $f_k \to f$ uniformly. It remains to show that f is continuous. Fix x, and choose $\delta > 0$ with $|f_N(y) - f_N(x)| < \varepsilon$ for $|y - x| < \delta$. Then for $|y - x| < \delta$,

$$|f(y) - f(x)| \le |f(y) - f_N(y)| + |f_N(y) - f_N(x)| + |f_N(x) - f(x)| < 3\varepsilon. \qquad \square$$

In particular, a uniform limit of continuous functions is continuous. A pointwise limit need not be: on $[0,1]$, x^k converges to the function that is 0 for $x < 1$ and 1 for $x = 1$.

Fixed points and differential equations. The proof of the Banach fixed-point theorem (Theorem A.5.3) only used the triangle inequality and completeness. So the theorem holds, with the same proof, for every contraction $T : C \to C$ of a closed nonempty subset C of a Banach space. This proves the existence theorem for differential equations of Lecture 3.

Theorem A.12.4 (Picard–Lindelöf)**.** *Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be Lipschitz continuous, $\|F(\mathbf{x}) - F(\mathbf{y})\| \le L\|\mathbf{x} - \mathbf{y}\|$, and let $T > 0$. Then for every $\mathbf{x}_0$, the ODE $\mathbf{x}' = F(\mathbf{x})$ has exactly one solution on $[0,T]$ with $\mathbf{x}(0) = \mathbf{x}_0$. Consequently, it has exactly one solution defined for all $t \in \mathbb{R}$ (Theorem 15.2 of Lecture 3).*

Proof. By the fundamental theorem of calculus, a continuous function $\mathbf{x} : [0,T] \to \mathbb{R}^n$ is a solution if and only if it is a fixed point of the map

$$(\Phi\mathbf{x})(t) = \mathbf{x}_0 + \int_0^t F(\mathbf{x}(s))\,ds,$$

which maps the space $C([0,T], \mathbb{R}^n)$ of continuous functions into itself. We give this space the weighted norm $\|\mathbf{x}\|_w = \max_{0 \le t \le T} e^{-2Lt}\|\mathbf{x}(t)\|$. Since $e^{-2LT} \le e^{-2Lt} \le 1$, it is equivalent to the maximum norm, and the space is complete, as in Proposition A.12.3. For continuous $\mathbf{x}$ and $\mathbf{y}$, and $0 \le t \le T$,

$$\|(\Phi\mathbf{x})(t) - (\Phi\mathbf{y})(t)\| \le \int_0^t L\|\mathbf{x}(s) - \mathbf{y}(s)\|\,ds \le \|\mathbf{x} - \mathbf{y}\|_w \int_0^t Le^{2Ls}\,ds \le \frac{e^{2Lt}}{2}\|\mathbf{x} - \mathbf{y}\|_w.$$

Multiplying by e^{-2Lt} gives $\|\Phi\mathbf{x} - \Phi\mathbf{y}\|_w \le \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|_w$. So Φ is a contraction, and it has exactly one fixed point. By uniqueness, the solutions on intervals $[0,T]$ of different lengths agree where they are both defined, so they form one solution on $[0,\infty)$. For negative times, apply the same argument to $\mathbf{y}(t) = \mathbf{x}(-t)$, which solves $\mathbf{y}' = -F(\mathbf{y})$. $\square$

With the maximum norm, the same estimate only shows that Φ is a contraction if $LT < 1$; the weight e^{-2Lt} removes this restriction. The iteration $\mathbf{x}_{k+1} = \Phi\mathbf{x}_k$ is called Picard iteration. For $x' = x$ with $x(0) = 1$, starting from the constant $x_0(t) = 1$, it gives $x_1(t) = 1 + t$, $x_2(t) = 1 + t + \frac{t^2}{2}$, and in general the Taylor polynomials of the solution e^t.

Bounded operators. Linear maps between infinite-dimensional spaces need not be continuous. A linear map $A : V \to W$ between normed spaces is bounded if its operator norm

$$\|A\| = \sup_{\|v\| \le 1} \|Av\|$$

is finite, and then $\|Av\| \le \|A\|\|v\|$ for all v, as for the matrix norm of Lecture 2. A linear map is continuous if and only if it is bounded: if it is bounded, then $\|Au - Av\| \le \|A\|\|u - v\|$; if it is continuous at 0, there is a $\delta > 0$ with $\|Av\| \le 1$ for $\|v\| \le \delta$, so $\|A\| \le \frac{1}{\delta}$. In finite dimensions, every linear map is bounded.

Example A.12.5.

1. (Integration.) The Volterra operator $(Vf)(x) = \int_0^x f(t)\,dt$ on $C[0,1]$ with the maximum norm satisfies $|(Vf)(x)| \le x\|f\|_\infty$, so $\|V\| = 1$, with equality for $f = 1$. By induction, $|(V^k f)(x)| \le \frac{x^k}{k!}\|f\|_\infty$, so $\|V^k\| = \frac{1}{k!}$: the powers of V decay much faster than $\|V\|^k = 1$.
2. (Differentiation.) On the polynomials on $[0,1]$ with the maximum norm, the derivative D is linear but not bounded: the polynomials x^k have the norm 1, but their derivatives kx^{k-1} have the norm k. Differentiation amplifies small wiggles of high frequency, which is why numerical differentiation by finite differences (Proposition 2.9 of Lecture 3) is sensitive to noise.

Dual spaces. The bounded linear functionals $\varphi : V \to \mathbb{R}$ on a normed space V form its dual space V^*, a Banach space with the operator norm. On $\mathbb{R}^n$, every linear functional is $\mathbf{x} \mapsto \mathbf{a}^\top \mathbf{x}$. The Hahn–Banach theorem, which we do not prove, guarantees that every normed space has many bounded linear functionals: for every closed subspace $U \ne V$, there is a $\varphi \in V^*$ with $\varphi \ne 0$ and $\varphi(u) = 0$ for all $u \in U$. The dual space of $C[a,b]$ consists of the integrals $f \mapsto \int f\,d\mu$ with respect to signed measures μ, i.e., differences of two finite measures on the Borel sets (the Riesz–Markov theorem). For example, the evaluation $f \mapsto f(x_0) = \int f\,d\delta_{x_0}$ at a point is a bounded linear functional on $C[a,b]$ with the norm 1. On $L^2[a,b]$, evaluation at a point is not even defined, since the elements of L^2 are only defined up to null sets. The dual of a Hilbert space is the space itself, as we show in the next section.

A.13 Hilbert Spaces

Definition A.13.1 (Hilbert space)**.** An inner product space is a real vector space H with an inner product $\langle \cdot, \cdot \rangle$ (Lecture 2) and the norm $\|v\| = \sqrt{\langle v, v \rangle}$. It is a Hilbert space if it is complete with this norm.

The Cauchy–Schwarz inequality $|\langle u, v \rangle| \le \|u\|\|v\|$ and Pythagoras' theorem hold as in Lecture 2, since their proofs only use the properties of an inner product, and so does the parallelogram law $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$ (expand both squares). By the Cauchy–Schwarz inequality, the inner product is continuous.

Example A.13.2 (Hilbert spaces)**.**

1. $\mathbb{R}^n$ with the dot product, and ℓ^2 with $\langle x, y \rangle = \sum_k x_k y_k$, which converges by the Cauchy–Schwarz inequality.
2. $L^2(\mu)$ with $\langle f, g \rangle = \int fg\,d\mu$, for example $L^2[a,b]$ with $\langle f, g \rangle = \int_a^b f(x)g(x)\,dx$. For a probability space, it is the space L^2 of random variables with finite variance and the inner product $\langle X, Y \rangle = \mathbb{E}[XY]$ of Lecture 4.

3. None of the other spaces ℓ^p is a Hilbert space, since their norms do not satisfy the parallelogram law; in ℓ^1, for example, the unit sequences $e_1 = (1, 0, 0, \ldots)$ and $e_2 = (0, 1, 0, \ldots)$ give $4 + 4 \neq 2 + 2$. And $C[0, 1]$ with the inner product of $L^2[0, 1]$ is an inner product space that is not complete, by the argument of Example A.12.2(4).

The most important property of Hilbert spaces is that we can project onto closed subspaces, as in Lecture 2.

Theorem A.13.3 (Projection theorem)**.** *Let C be a nonempty closed convex subset of a Hilbert space H. Then every $x \in H$ has exactly one closest point $P_C x$ in C. If $C = U$ is a closed subspace, then $P_U x$ is the only $p \in U$ with $x - p \perp U$, and every $x \in H$ can be written in exactly one way as $x = u + w$ with $u \in U$ and $w \perp U$.*

Proof. Let $d = \inf_{c \in C} \|x - c\|$, and choose $c_k \in C$ with $\|x - c_k\| \to d$. The parallelogram law for $x - c_k$ and $x - c_l$ gives

$$\|c_k - c_l\|^2 = 2\|x - c_k\|^2 + 2\|x - c_l\|^2 - 4\left\|x - \frac{c_k + c_l}{2}\right\|^2 \leq 2\|x - c_k\|^2 + 2\|x - c_l\|^2 - 4d^2,$$

since $\frac{c_k + c_l}{2} \in C$ by convexity. The right side tends to 0 as $k, l \to \infty$, so (c_k) is a Cauchy sequence. Its limit c lies in C, since C is closed, and $\|x - c\| = d$ by the continuity of the norm. If c' is another closest point, the same inequality for c and c' gives $\|c - c'\|^2 \leq 0$. Now let $C = U$ be a closed subspace, and $p = P_U x$. For $u \in U$, the function $t \mapsto \|x - p - tu\|^2 = \|x - p\|^2 - 2t\langle x - p, u\rangle + t^2\|u\|^2$ is minimal at $t = 0$, so $\langle x - p, u\rangle = 0$. Conversely, if $p' \in U$ and $x - p' \perp U$, then $\|x - u\|^2 = \|x - p'\|^2 + \|p' - u\|^2$ for all $u \in U$ by Pythagoras, so p' is the closest point. Finally, $x = P_U x + (x - P_U x)$, and this decomposition is unique since $U \cap U^\perp = \{0\}$. □

For a subspace U of $\mathbb{R}^n$, this is the least squares projection of Lecture 2. The subspace must be closed: the polynomials form a subspace of $L^2[0, 1]$ that is not closed, and the function e^x has the distance 0 from it (by the Weierstrass approximation theorem, Section A.16), but no closest polynomial. One can show that the random variables in L^2 that are measurable with respect to a σ-algebra $\mathcal{G}$ form a closed subspace, and the projection onto it is the conditional expectation $\mathbb{E}[Y \mid \mathcal{G}]$ (Theorem B.7.2 of the appendix to Lecture 4).

Theorem A.13.4 (Riesz representation)**.** *For every bounded linear functional φ on a Hilbert space H, there is exactly one $y \in H$ with $\varphi(x) = \langle x, y\rangle$ for all $x \in H$, and $\|\varphi\| = \|y\|$.*

Proof. If $\varphi = 0$, take $y = 0$. Otherwise, the kernel $U = \{x : \varphi(x) = 0\}$ is a closed subspace (the preimage of $\{0\}$ under a continuous map), and $U \neq H$. By the projection theorem, there is a $z \neq 0$ with $z \perp U$ (take $z = v - P_U v$ for some $v \notin U$), and we may scale it so that $\varphi(z) = 1$. For every x, the vector $x - \varphi(x)z$ lies in U, so it is orthogonal to z, i.e., $\langle x, z\rangle = \varphi(x)\|z\|^2$. So $y = \frac{z}{\|z\|^2}$ has the required property. If y' has it too, then $\langle x, y - y'\rangle = 0$ for all x, in particular for $x = y - y'$, so $y = y'$. Finally, $|\varphi(x)| \leq \|x\|\|y\|$ by the Cauchy–Schwarz inequality, with equality for $x = y$. □

The representing vector depends on the inner product. This is why the gradient depends on the inner product (Remark 3.8 of Lecture 3): the derivative $Df(\mathbf{x})$ is a linear functional, and the gradient is its Riesz representer. The Riesz representation theorem also proves the Radon–Nikodym theorem of the appendix to Lecture 4, by an argument of von Neumann.

Theorem A.13.5 (Radon–Nikodym)**.** *Let μ and ν be finite measures on a measurable space $(\Omega, \mathcal{F})$ such that $\mu(A) = 0$ implies $\nu(A) = 0$. Then there is a measurable $p \geq 0$ with $\nu(A) = \int_A p \, d\mu$ for all $A \in \mathcal{F}$.*

Sketch of proof. Let $\rho = \mu + \nu$. For $f \in L^2(\rho)$, the Cauchy–Schwarz inequality gives $|\int f\,d\nu| \le \int |f|\,d\rho \le \rho(\Omega)^{1/2}\|f\|_{L^2(\rho)}$, so $f \mapsto \int f\,d\nu$ is a bounded linear functional on $L^2(\rho)$. By the Riesz representation theorem, there is a $g \in L^2(\rho)$ with

$$\int f\,d\nu = \int fg\,d\rho = \int fg\,d\mu + \int fg\,d\nu \qquad \text{for all } f \in L^2(\rho).$$

For $f = \mathbf{1}_A$, this gives $\nu(A) = \int_A g\,d\rho$, and since $0 \le \nu(A) \le \rho(A)$ for all A, one shows that $0 \le g \le 1$ almost everywhere. Rearranging, $\int f(1-g)\,d\nu = \int fg\,d\mu$. For $f = \mathbf{1}_N$ with $N = \{g = 1\}$, the left side is 0, so $\mu(N) = 0$, and hence also $\nu(N) = 0$. Now let $A \in \mathcal{F}$, and take $f = \mathbf{1}_A(1 + g + \cdots + g^k)$, so that $f(1-g) = \mathbf{1}_A(1 - g^{k+1})$:

$$\int_A (1 - g^{k+1})\,d\nu = \int_A (g + g^2 + \cdots + g^{k+1})\,d\mu.$$

Outside the null set N, $g^{k+1} \to 0$ as $k \to \infty$, and the monotone convergence theorem (Theorem B.4.1 of the appendix to Lecture 4) on both sides gives $\nu(A) = \int_A \frac{g}{1-g}\,d\mu$. So $p = \frac{g}{1-g}$ is the density. □

This is Theorem B.5.2 of the appendix to Lecture 4 for finite measures; the σ-finite case follows by splitting Ω into countably many pieces of finite measure.

Orthonormal bases. In infinite dimensions, a basis in the sense of Lecture 1, in which every vector is a finite linear combination of basis vectors, is not useful. Instead, we allow infinite sums.

Theorem A.13.6 (Orthonormal bases)**.** *Let $e_1, e_2, \ldots$ be orthonormal vectors in a Hilbert space H, and let $x \in H$.*

1. *(Bessel's inequality) $\sum_k \langle x, e_k\rangle^2 \le \|x\|^2$.*
2. *The series $\sum_k \langle x, e_k\rangle e_k$ converges, and its sum is the orthogonal projection of x onto the closure U of* $\operatorname{span}\{e_1, e_2, \ldots\}$.
3. *If $U = H$, then $x = \sum_k \langle x, e_k\rangle e_k$ and $\|x\|^2 = \sum_k \langle x, e_k\rangle^2$ (Parseval's identity) for every $x \in H$. Then $e_1, e_2, \ldots$ is called an orthonormal basis of H.*

Proof. Let $c_k = \langle x, e_k\rangle$ and $s_n = \sum_{k\le n} c_k e_k$. Then $x - s_n \perp e_1, \ldots, e_n$, so Pythagoras gives $\|x\|^2 = \|x - s_n\|^2 + \sum_{k\le n} c_k^2$, which proves (1). For $m > n$, $\|s_m - s_n\|^2 = \sum_{k=n+1}^m c_k^2$, which tends to 0 as $n \to \infty$ by (1). So (s_n) is a Cauchy sequence, with a limit $s \in U$. By the continuity of the inner product, $\langle x - s, e_k\rangle = \lim_n \langle x - s_n, e_k\rangle = 0$ for all k, so $x - s$ is orthogonal to the span of the e_k, and, again by continuity, to its closure U. By Theorem A.13.3, $s = P_U x$. If $U = H$, then $s = x$, so $\|x - s_n\| \to 0$, and the identity $\|x\|^2 = \|x - s_n\|^2 + \sum_{k\le n} c_k^2$ gives Parseval's identity in the limit. □

So in a Hilbert space with an orthonormal basis, a vector behaves like the sequence of its coefficients: the map $x \mapsto (\langle x, e_k\rangle)_k$ is a linear bijection from H onto ℓ^2 that preserves inner products. One can show that every Hilbert space with a countable dense subset, such as $L^2[a, b]$, has an orthonormal basis, by the Gram–Schmidt process of Lecture 2 applied to a dense sequence. So all these spaces of infinite dimension are copies of ℓ^2. For example, the Gram–Schmidt process applied to $1, x, x^2, \ldots$ in $L^2[-1, 1]$ gives the normalized Legendre polynomials $\frac{1}{\sqrt{2}}$, $\sqrt{\frac{3}{2}}\,x$, $\sqrt{\frac{5}{8}}\,(3x^2 - 1)$, …, which form an orthonormal basis, since the polynomials are dense in $L^2[-1, 1]$.

Example A.13.7 (Fourier series)**.** The functions $\frac{1}{\sqrt{2\pi}}$, $\frac{1}{\sqrt{\pi}}\cos kx$ and $\frac{1}{\sqrt{\pi}}\sin kx$ for $k = 1, 2, \ldots$ are orthonormal in $L^2[-\pi, \pi]$, by the product formulas for sines and cosines, and one can show that they form an orthonormal basis. The expansion of a function f in this basis is its Fourier series,

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} \big(a_k \cos kx + b_k \sin kx\big), \qquad a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos kx\, dx, \quad b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin kx\, dx,$$

which converges in $L^2[-\pi, \pi]$. For $f(x) = x$, integration by parts gives $a_k = 0$ and $b_k = \frac{2(-1)^{k+1}}{k}$, so

$$x = 2\Big(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \cdots\Big) \qquad \text{in } L^2[-\pi, \pi].$$

The coefficients with respect to the orthonormal functions $\frac{1}{\sqrt{\pi}}\sin kx$ are $\sqrt{\pi}\, b_k = \frac{2\sqrt{\pi}(-1)^{k+1}}{k}$, so Parseval's identity gives $\int_{-\pi}^{\pi} x^2\, dx = \frac{2\pi^3}{3} = \sum_k \frac{4\pi}{k^2}$, i.e.,

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

The partial sums converge to f in L^2, but not uniformly: near the jumps of the periodic extension of f at $\pm\pi$, they overshoot by about 9% of the jump, however many terms we take (the Gibbs phenomenon). The Fourier coefficients of smooth periodic functions decay quickly, so that a few of them describe the function well. Compression methods for sound and images, such as MP3 and JPEG, are based on this idea, with related bases.

A.14 Compact Operators and the Spectral Theorem

The spectral theorem of Lecture 2 says that a symmetric matrix has an orthonormal basis of eigenvectors. In infinite dimensions, a bounded symmetric operator need not have any eigenvectors: the multiplication operator $(Mf)(x) = xf(x)$ on $L^2[0, 1]$ satisfies $\langle Mf, g\rangle = \langle f, Mg\rangle$, but $Mf = \lambda f$ forces $f = 0$ outside the null set $\{\lambda\}$, i.e., $f = 0$ in L^2. The spectral theorem holds for operators that are close to matrices in the following sense.

Definition A.14.1 (Compact and self-adjoint operators)**.** A bounded linear operator $K : H \to H$ on a Hilbert space is compact if for every bounded sequence (x_k) in H, the sequence (Kx_k) has a convergent subsequence. It is self-adjoint if $\langle Kx, y\rangle = \langle x, Ky\rangle$ for all $x, y \in H$, and positive semidefinite if moreover $\langle Kx, x\rangle \geq 0$ for all x.

By the Bolzano–Weierstrass theorem (Theorem A.3.1), every bounded operator whose image is finite-dimensional is compact. The identity of an infinite-dimensional Hilbert space is not compact (Remark A.3.11).

Theorem A.14.2 (Spectral theorem for compact operators)**.** *Let K be a compact self-adjoint operator on a Hilbert space H. Then there are finitely or countably many orthonormal eigenvectors $e_1, e_2, \ldots$ of K with nonzero eigenvalues $\lambda_1, \lambda_2, \ldots$, ordered so that $|\lambda_1| \geq |\lambda_2| \geq \cdots$, such that*

$$Kx = \sum_k \lambda_k \langle x, e_k\rangle e_k \qquad \textit{for all } x \in H.$$

If there are infinitely many, then $\lambda_k \to 0$, and the eigenspace of each nonzero eigenvalue is finite-dimensional. If K is positive semidefinite, all λ_k are positive.

Idea of proof. One shows that $|\langle Kx, x\rangle|$ attains its maximum on the unit ball, by compactness (this replaces the compactness of the unit sphere in $\mathbb{R}^n$), and that a maximizer is an eigenvector with the eigenvalue $\pm\|K\|$, as for the Rayleigh quotient in Lecture 2. Then one repeats this on the orthogonal complement of the eigenvectors found so far, which K maps into itself. If there were infinitely many orthonormal eigenvectors with $|\lambda_k| \geq \varepsilon > 0$, then $\|Ke_k - Ke_l\|^2 = \lambda_k^2 + \lambda_l^2 \geq 2\varepsilon^2$ for $k \neq l$, so the sequence (Ke_k) would have no convergent subsequence, although (e_k) is bounded. This shows that $\lambda_k \to 0$, and that the eigenspaces of nonzero eigenvalues are finite-dimensional. □

Integral operators. Let $k : [a,b]^2 \to \mathbb{R}$ be continuous and symmetric, $k(x,y) = k(y,x)$. The integral operator

$$(Kf)(x) = \int_a^b k(x,y) f(y)\, dy$$

is the infinite-dimensional analogue of the product of a symmetric matrix with a vector, with $k(x,y)$ in the role of the entries. By Fubini's theorem, it is self-adjoint on $L^2[a,b]$, and one can show that it is compact. We call k positive semidefinite if the matrix $(k(x_i,x_j))_{i,j}$ is positive semidefinite for all points $x_1, \ldots, x_n \in [a,b]$, as for the kernels of Gaussian processes in Lecture 4; then K is positive semidefinite.

Theorem A.14.3 (Mercer)**.** *Let k be continuous, symmetric and positive semidefinite on $[a,b]^2$, and let λ_k and e_k be the positive eigenvalues and the orthonormal eigenfunctions of K. Then the e_k are continuous, and*

$$k(x,y) = \sum_k \lambda_k e_k(x) e_k(y),$$

with absolute and uniform convergence on $[a,b]^2$. In particular, $\sum_k \lambda_k = \int_a^b k(x,x)\, dx$.

We do not prove Mercer's theorem. It says that $k(x,y) = \langle \boldsymbol{\phi}(x), \boldsymbol{\phi}(y)\rangle$ is the inner product in ℓ^2 of the features $\boldsymbol{\phi}(x) = (\sqrt{\lambda_k}\, e_k(x))_k$: every continuous positive semidefinite kernel is an inner product of (possibly infinitely many) features, as the kernels of the kernel trick in Lecture 2.

Example A.14.4 (The Karhunen–Loève expansion of Brownian motion)**.** The covariance function $k(s,t) = \min(s,t)$ of Brownian motion on $[0,1]$ (Proposition 15.2 of Lecture 4) is continuous, symmetric and positive semidefinite, since covariance matrices are positive semidefinite. Its integral operator is

$$(Kf)(t) = \int_0^t s f(s)\, ds + t \int_t^1 f(s)\, ds.$$

Differentiating, $(Kf)'(t) = tf(t) + \int_t^1 f(s)\, ds - tf(t) = \int_t^1 f(s)\, ds$, so $(Kf)'' = -f$, $(Kf)(0) = 0$ and $(Kf)'(1) = 0$. Hence an eigenfunction with $Ke = \lambda e$ and $\lambda > 0$ satisfies $\lambda e'' = -e$ with $e(0) = 0$ and $e'(1) = 0$. So $e(t)$ is a multiple of $\sin(t/\sqrt{\lambda})$ with $\cos(1/\sqrt{\lambda}) = 0$, which gives the normalized eigenfunctions and the eigenvalues

$$e_k(t) = \sqrt{2}\sin\big((k - \tfrac{1}{2})\pi t\big), \qquad \lambda_k = \frac{1}{(k-\frac{1}{2})^2\pi^2}, \qquad k = 1, 2, \ldots$$

As Mercer's theorem predicts, $\sum_k \lambda_k = \frac{4}{\pi^2}\sum_k \frac{1}{(2k-1)^2} = \frac{4}{\pi^2}\cdot\frac{\pi^2}{8} = \frac{1}{2} = \int_0^1 \min(t,t)\, dt$, where the sum over the odd numbers is $\frac{\pi^2}{6} - \frac{1}{4}\cdot\frac{\pi^2}{6}$, by Example A.13.7. Now let $Z_1, Z_2, \ldots$ be i.i.d. $\mathcal{N}(0,1)$, and

$$B(t) = \sum_{k=1}^{\infty} \sqrt{\lambda_k}\, Z_k e_k(t).$$

By independence and Mercer's theorem, $\mathbb{E}[B(s)B(t)] = \sum_k \lambda_k e_k(s)e_k(t) = \min(s,t)$: B has the covariance function of Brownian motion, and one can show that the series converges and defines a Brownian motion. This is the Karhunen–Loève expansion. It writes a random function as a series with independent coefficients, ordered by their variances: it is the principal component analysis (Lecture 2) of a random function. The first term carries the fraction $\frac{\lambda_1}{1/2} = \frac{8}{\pi^2} \approx 81\%$ of the total variance $\mathbb{E}\big[\int_0^1 B(t)^2\,dt\big] = \frac{1}{2}$, and the first three terms carry 93%.

A.15 Reproducing Kernel Hilbert Spaces

Kernel methods and Gaussian processes (Lecture 4) work with functions through a kernel. The natural Hilbert spaces for them are spaces of functions in which evaluating a function at a point is continuous, unlike in L^2.

Definition A.15.1 (Reproducing kernel Hilbert space)**.** Let X be a set. A Hilbert space H of functions $f : X \to \mathbb{R}$ is a reproducing kernel Hilbert space (RKHS) if for every $x \in X$, the evaluation $f \mapsto f(x)$ is a bounded linear functional on H.

By the Riesz representation theorem, for every $x \in X$ there is a function $k_x \in H$ with

$$f(x) = \langle f, k_x \rangle \qquad \text{for all } f \in H,$$

the reproducing property. The function $k(x,y) = \langle k_x, k_y \rangle = k_y(x)$ is called the reproducing kernel of H. It is symmetric, and it is positive semidefinite: for points $x_1, \dots, x_n$ and $\mathbf{a} \in \mathbb{R}^n$,

$$\sum_{i,j} a_i a_j k(x_i, x_j) = \Big\| \sum_i a_i k_{x_i} \Big\|^2 \geq 0.$$

Conversely, by the Moore–Aronszajn theorem, which we do not prove, every symmetric positive semidefinite kernel k on X is the reproducing kernel of exactly one RKHS H_k. It is the completion of the space of the finite combinations $\sum_i a_i k_{x_i}$ with the inner product $\langle \sum_i a_i k_{x_i}, \sum_j b_j k_{y_j} \rangle = \sum_{i,j} a_i b_j k(x_i, y_j)$. In particular, $\|\sum_i a_i k_{x_i}\|^2 = \mathbf{a}^\top K \mathbf{a}$, with the kernel matrix $K_{ij} = k(x_i, x_j)$.

Example A.15.2.

1. (Linear kernel.) For $k(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{y}$ on $\mathbb{R}^d$, H_k consists of the linear functions $f_{\mathbf{w}}(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$, with $\langle f_{\mathbf{v}}, f_{\mathbf{w}} \rangle = \mathbf{v}^\top \mathbf{w}$: indeed, $k_{\mathbf{x}} = f_{\mathbf{x}}$, and $\langle f_{\mathbf{w}}, k_{\mathbf{x}} \rangle = \mathbf{w}^\top \mathbf{x} = f_{\mathbf{w}}(\mathbf{x})$. So the RKHS norm $\|f_{\mathbf{w}}\| = \|\mathbf{w}\|$ of a linear function is the norm of its weight vector, which ridge regression penalizes.

2. (Feature maps.) Similarly, for $k(x,y) = \boldsymbol{\phi}(x)^\top \boldsymbol{\phi}(y)$ with a feature map $\boldsymbol{\phi} : X \to \mathbb{R}^m$, H_k consists of the functions $f_{\mathbf{w}}(x) = \mathbf{w}^\top \boldsymbol{\phi}(x)$, and $\|f\|$ is the smallest norm of a $\mathbf{w}$ with $f = f_{\mathbf{w}}$. For a kernel as in Mercer's theorem, with the features $\sqrt{\lambda_k}\, e_k(x)$, H_k consists of the functions $f = \sum_k c_k e_k$ with $\|f\|^2 = \sum_k \frac{c_k^2}{\lambda_k} < \infty$. Since $\lambda_k \to 0$, the coefficients of the functions in H_k decay quickly.

3. (Smoothness.) For the kernel $\min(s,t)$ on $[0,1]$, H_k consists of the functions $f(t) = \int_0^t g(s)\,ds$ with $g \in L^2[0,1]$, with $\langle f_1, f_2 \rangle = \int_0^1 f_1'(s) f_2'(s)\,ds$. Indeed, $k_s(t) = \min(s,t)$ has the derivative $\mathbf{1}_{[0,s]}$, so $\langle f, k_s \rangle = \int_0^s f'(u)\,du = f(s)$. The RKHS norm $\|f\| = \|f'\|_{L^2}$ measures how much f varies. Similarly, one can show that all functions in the RKHS of the squared exponential kernel are infinitely differentiable.

The RKHS norm is a natural regularizer for learning a function from data, and the following theorem makes such learning problems finite-dimensional.

Theorem A.15.3 (Representer theorem)**.** *Let H be an RKHS on X with the kernel k, and let $\lambda > 0$. For any points $x_1, \dots, x_N$ in X and any function $L : \mathbb{R}^N \to \mathbb{R}$, every minimizer f over H of*

$$L(f(x_1), \dots, f(x_N)) + \lambda \|f\|^2$$

has the form $f = \sum_{i=1}^N \alpha_i k_{x_i}$ with $\alpha_1, \dots, \alpha_N \in \mathbb{R}$.

Proof. The span U of $k_{x_1}, \dots, k_{x_N}$ is finite-dimensional, hence closed. Write $f = u + w$ with $u \in U$ and $w \perp U$ (Theorem A.13.3). Then $f(x_i) = \langle f, k_{x_i} \rangle = \langle u, k_{x_i} \rangle = u(x_i)$, so the first term is the same for f and u, while $\|f\|^2 = \|u\|^2 + \|w\|^2$. If $w \neq 0$, then u has a smaller objective than f, so f is not a minimizer. □

So an optimization problem over an infinite-dimensional space of functions reduces to the N coefficients α_i.

Example A.15.4 (Kernel ridge regression)**.** For the squared loss, we minimize $\sum_{i=1}^N (y_i - f(x_i))^2 + \lambda \|f\|^2$. By the representer theorem, $f = \sum_j \alpha_j k_{x_j}$, so $f(x_i) = (K\boldsymbol{\alpha})_i$ and $\|f\|^2 = \boldsymbol{\alpha}^\top K \boldsymbol{\alpha}$, and we have to minimize $\|\mathbf{y} - K\boldsymbol{\alpha}\|^2 + \lambda \boldsymbol{\alpha}^\top K \boldsymbol{\alpha}$ over $\boldsymbol{\alpha} \in \mathbb{R}^N$. This convex function has the gradient $2K((K + \lambda I)\boldsymbol{\alpha} - \mathbf{y})$, which vanishes for $\boldsymbol{\alpha} = (K + \lambda I)^{-1}\mathbf{y}$. So the prediction at a new point x_* is

$$f(x_*) = \mathbf{k}_*^\top (K + \lambda I)^{-1} \mathbf{y}, \qquad \mathbf{k}_* = (k(x_1, x_*), \dots, k(x_N, x_*))^\top,$$

the posterior mean of Gaussian process regression with the noise variance $\sigma^2 = \lambda$ (Proposition 12.11 of Lecture 4). For the data of Example 12.12 of Lecture 4, with $\lambda = \sigma^2 = 0.1$, the coefficients are $\boldsymbol{\alpha} = (1.306, -0.720)$, the prediction at $x_* = 0.5$ is $f(0.5) = 0.517$, and the RKHS norm of the regression function is given by $\|f\|^2 = \boldsymbol{\alpha}^\top K \boldsymbol{\alpha} \approx 1.084$.

A.16 Functional Analysis in Machine Learning

- **Universal approximation.** By the Weierstrass approximation theorem, which we do not prove, the polynomials are dense in $C[a, b]$: every continuous function is a uniform limit of polynomials. Since $\|f\|_{L^2} \leq \sqrt{b - a}\, \|f\|_\infty$, and one can show that the continuous functions are dense in $L^2[a, b]$, the polynomials are also dense in $L^2[a, b]$. Neural networks have the same property. Cybenko's theorem (1989) says that for a continuous function σ with $\sigma(t) \to 0$ as $t \to -\infty$ and $\sigma(t) \to 1$ as $t \to \infty$, such as the logistic sigmoid, the networks $\sum_{i=1}^m c_i \sigma(\mathbf{w}_i^\top \mathbf{x} + b_i)$ with one hidden layer are dense in $C(Q)$, the continuous functions on a box $Q \subseteq \mathbb{R}^n$ with the maximum norm. The proof is functional analysis. If the closure U of the subspace of these networks were not all of $C(Q)$, the Hahn–Banach theorem would give a nonzero bounded linear functional that vanishes on U, and by the Riesz–Markov theorem, it would be a signed measure $\mu \neq 0$ with $\int \sigma(\mathbf{w}^\top \mathbf{x} + b)\, d\mu(\mathbf{x}) = 0$ for all $\mathbf{w}$ and b. One then shows that this forces $\mu = 0$. The theorem says nothing about how many neurons are needed, or whether training finds good weights.

- **Kernel methods.** Support vector machines, kernel ridge regression and Gaussian processes work in the RKHS of a kernel, which is often infinite-dimensional. By the representer theorem, all computations only involve the $N \times N$ kernel matrix: this is the kernel trick of Lecture 2. The RKHS norm is the regularizer, and for kernels such as the squared exponential kernel, it penalizes functions that are not smooth.

- **Gaussian processes and their RKHS.** The posterior mean of a Gaussian process lies in the RKHS of its kernel, but the paths of the process are rougher than the functions in the RKHS. For Brownian motion, the RKHS consists of the functions with square-integrable derivatives (Example A.15.2), while the paths are nowhere differentiable (Section 15 of Lecture 4). In terms of the Karhunen–Loève expansion $B = \sum_k \sqrt{\lambda_k} Z_k e_k$, the RKHS norm of a path would be $\sum_k Z_k^2$, which is infinite almost surely, by the strong law of large numbers (Theorem B.9.6 of the appendix to Lecture 4).

- **The neural tangent kernel.** For a network $f(\mathbf{x}; \boldsymbol{\theta})$, the kernel

$$\Theta(\mathbf{x}, \mathbf{x}') = \nabla_{\boldsymbol{\theta}} f(\mathbf{x}; \boldsymbol{\theta})^\top \nabla_{\boldsymbol{\theta}} f(\mathbf{x}'; \boldsymbol{\theta})$$

  describes how a gradient step on one example changes the prediction at another input. One can show that for infinitely wide networks with a suitably scaled random initialization, this kernel stays constant during training, and that training by gradient descent on the squared loss then amounts to kernel regression with this kernel. This connects deep learning with the kernel methods of this appendix, at least in this limit.

- **Functional gradients and boosting.** Gradient boosting performs gradient descent in a space of functions: at each step, it fits a simple model, such as a small decision tree, to the negative gradient of the loss with respect to the predictions $f(\mathbf{x}_i)$, and adds a small multiple of it to the current model.

- **Neural operators.** Some learning problems map functions to functions, for example the initial condition of a differential equation to the solution at a later time. Neural operators learn such maps between infinite-dimensional spaces, for example by acting on the Fourier coefficients of their inputs (Example A.13.7).

Key Takeaways

- Open sets, closed sets, convergence and continuity only depend on the notion of nearness. A map is continuous if and only if preimages of open sets are open; so sublevel sets $\{f \leq c\}$ and level sets of continuous functions are closed. All norms on $\mathbb{R}^n$ define the same open sets.

- A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded (Heine–Borel). A continuous function on a nonempty compact set attains its minimum and maximum, and a continuous coercive function on $\mathbb{R}^n$ has a global minimum; regularization makes losses coercive.

- Continuous maps preserve connectedness; this gives the intermediate value theorem, and shows, e.g., that $\mathrm{GL}(n, \mathbb{R})$ has two pieces and that a circle is not homeomorphic to an interval.

- A contraction of a closed set has a unique fixed point, found by iteration (Banach). This proves the inverse function theorem: a map with an invertible Jacobian is locally invertible, with $D(\mathbf{f}^{-1}) = (D\mathbf{f})^{-1}$. Residual layers $\mathbf{x} + \mathbf{g}(\mathbf{x})$ with a contractive $\mathbf{g}$ are invertible.

- The implicit function theorem solves $F(\mathbf{x}, \mathbf{y}) = \mathbf{0}$ locally for $\mathbf{y} = \mathbf{g}(\mathbf{x})$ when $D_{\mathbf{y}} F$ is invertible, with $D\mathbf{g} = -D_{\mathbf{y}} F^{-1} D_{\mathbf{x}} F$; this is implicit differentiation, also used for implicit layers and hyperparameter optimization.

- The level set of a regular value of $F : \mathbb{R}^n \to \mathbb{R}^m$ is an $(n-m)$-dimensional submanifold, which is locally a graph. Its tangent space is $\operatorname{null} DF(\mathbf{p})$, and its normal space is spanned by the gradients $\nabla F_i(\mathbf{p})$. Examples: spheres, $\mathrm{SL}(n)$ (dimension n^2-1, tangent space at I: traceless matrices), $\mathrm{O}(n)$ (dimension $\frac{n(n-1)}{2}$, tangent space at I: skew-symmetric matrices).

- Lagrange multipliers say that ∇f is normal to the constraint manifold at a constrained extremum. The Riemannian gradient is the projection of ∇f onto the tangent space; Riemannian gradient descent alternates such steps with retractions.

- By Sard's theorem, almost all values of a smooth map are regular. A critical point is nondegenerate if its Hessian is invertible, and its index is the number of negative eigenvalues. Nondegenerate critical points are isolated, and by the Morse lemma f is exactly a sum of $\pm y_i^2$ near them. Almost every linear perturbation of a function is a Morse function.

- Sublevel sets change their topology only at critical values: a critical point of index k attaches a k-cell. On a compact manifold, $\chi(M) = \sum_k (-1)^k c_k$.

- Alternating k-forms on $\mathbb{R}^n$ have the basis $dx_I = dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ with $i_1 < \cdots < i_k$, so $\dim \Lambda^k(\mathbb{R}^n)^* = \binom{n}{k}$, and $(\varphi_1 \wedge \cdots \wedge \varphi_k)(v_1, \ldots, v_k) = \det(\varphi_i(v_j))$. The exterior derivative of $\sum_I f_I \, dx_I$ is $\sum_I df_I \wedge dx_I$, with $d \circ d = 0$; in $\mathbb{R}^3$, it is the gradient, the curl and the divergence. Pullbacks of top-degree forms produce Jacobian determinants, and integrals of forms over k-cubes do not depend on the parametrization (up to orientation). Forms are integrated over compact oriented manifolds with boundary with local parametrizations and partitions of unity. Stokes' theorem $\int_M d\omega = \int_{\partial M} \omega$, with the orientation "outward normal first" on ∂M, contains all the integral theorems of Lecture 3; it is proved for cubes with Fubini's theorem and the fundamental theorem of calculus, and the general case reduces to cubes by pullbacks. Closed forms that are not exact, such as the angle form on $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ and the solid angle form on $\mathbb{R}^3 \setminus \{\mathbf{0}\}$, detect holes; the de Rham cohomology spaces H^k count them.

- A Banach space is a complete normed space, such as $C[a,b]$ with the maximum norm, ℓ^p and L^p. The Banach fixed-point theorem holds in it; with a weighted maximum norm, it proves the existence and uniqueness of solutions of ODEs (Picard–Lindelöf). A linear map is continuous if and only if it is bounded; integration is bounded, differentiation is not.

- In a Hilbert space, every closed convex set has a unique closest point (projection theorem), every bounded linear functional is an inner product with a fixed vector (Riesz), and an orthonormal basis gives the expansion $x = \sum_k \langle x, e_k \rangle e_k$ with Parseval's identity, such as Fourier series. Conditional expectations are projections in L^2, and the Riesz representation theorem proves the Radon–Nikodym theorem.

- A compact self-adjoint operator has orthonormal eigenvectors with eigenvalues that tend to 0, and it is diagonal in them. For a continuous positive semidefinite kernel, Mercer's theorem gives $k(x,y) = \sum_k \lambda_k e_k(x) e_k(y)$, and the Karhunen–Loève expansion of a random function, such as Brownian motion, is its principal component analysis.

- In an RKHS, evaluation is continuous, $f(x) = \langle f, k_x \rangle$, and every positive semidefinite kernel has one. By the representer theorem, regularized learning in an RKHS reduces to N coefficients; kernel ridge regression gives $\boldsymbol{\alpha} = (K + \lambda I)^{-1}\mathbf{y}$, the posterior mean of a Gaussian process. Networks with one hidden layer are dense in the continuous functions (Cybenko).

Exercises

1. Which of the following subsets of $\mathbb{R}^2$ are open, closed, both or neither? Find their interior, closure and boundary: (a) $\{x > 0, y > 0\}$; (b) $\{x^2 + y^2 \leq 1\}$; (c) $\{(x, 0) : 0 \leq x < 1\}$; (d) $\mathbb{Q}^2$, the points with rational coordinates.

2. Show that the union of infinitely many closed sets need not be closed. Show that a set $U \subseteq X$ is open if and only if it is a union of open balls.

3. (a) Show that the set of $m \times n$ matrices of rank at most r is closed in $\mathbb{R}^{m\times n}$. (Hint: the rank is at most r if and only if all $(r+1) \times (r+1)$ minors are 0.)

   (b) Show that every square matrix A is a limit of invertible matrices. (Hint: $\det(A + tI)$ is a polynomial in t.) So $\mathrm{GL}(n, \mathbb{R})$ is open and dense in $\mathbb{R}^{n\times n}$.

4. Which of the following sets are compact? (a) The closed unit ball of the ℓ_1 norm in $\mathbb{R}^n$; (b) $\{(x, y) : xy = 1\}$; (c) $\mathrm{SO}(n)$; (d) $\mathrm{GL}(n, \mathbb{R})$; (e) the ellipsoid $\{\mathbf{x} : \mathbf{x}^\top A\mathbf{x} \leq 1\}$ for a symmetric positive definite A; (f) the same set for $A = \mathrm{diag}(1, -1)$.

5. Show that the regularized logistic regression loss $\frac{1}{N}\sum_i \log(1 + e^{-y_i \mathbf{w}^\top \mathbf{x}_i}) + \lambda\|\mathbf{w}\|^2$ with $\lambda > 0$ has a global minimum, and that it is unique. (Use Proposition A.3.6 and Lecture 3.)

6. Let $A \in \mathbb{R}^{n\times n}$. Show that $N(\mathbf{x}) = \|A\mathbf{x}\|$ is a norm if and only if A is invertible, and that the best constants in $c\|\mathbf{x}\| \leq N(\mathbf{x}) \leq C\|\mathbf{x}\|$ are the smallest and the largest singular value of A (Lecture 2).

7. (a) Show that the unit circle S^1 is path-connected, and that the hyperbola $\{x^2 - y^2 = 1\}$ is not connected.

   (b) Show that $\mathrm{SO}(2)$ is path-connected, and that $\mathrm{O}(2)$ is not connected.

   (c) Show that no continuous injective map from S^1 to $\mathbb{R}$ exists. (Hint: intermediate value theorem.)

8. Use the iteration $x_{k+1} = \cos x_k$ with $x_0 = 1$ to compute the solution of $\cos x = x$ to six decimal places. How many steps does the a priori bound of Theorem A.5.3 require, and how many do you actually need?

9. Show that $f(x) = x + \frac{1}{2}\tanh(x)$ is a homeomorphism of $\mathbb{R}$, and compute $f^{-1}(1)$ with the iteration of Corollary A.5.5, starting at $x_0 = 0$. Is f a diffeomorphism?

10. At which points of the folium $x^3 + y^3 = 3xy$ can the curve not be written locally as a graph $y = g(x)$? Compute $g'(x)$ at the point $(\frac{3}{2}, \frac{3}{2})$.

11. Let $F(x, y, z) = x^2 + y^2 - z^2$. Find all critical points and critical values of F. For which c is $F^{-1}(c)$ a submanifold? Describe the level sets for $c > 0$, $c = 0$ and $c < 0$; which of them are connected?

12. (a) Find the tangent space of the sphere $x^2+y^2+z^2 = 9$ at $(1, 2, 2)$ as $\operatorname{null} DF$, and the tangent plane $(1, 2, 2) + T_{(1,2,2)}M$ as an equation.

    (b) Show that $T_Q\mathrm{O}(n) = \{QK : K^\top = -K\}$ and $T_I\mathrm{SL}(n, \mathbb{R}) = \{H : \operatorname{tr} H = 0\}$ have the dimensions predicted by Example A.7.3.

13. (Orthogonal Procrustes problem.) Given $A \in \mathbb{R}^{n\times n}$ with SVD $A = U\Sigma V^\top$, show that $\operatorname{tr}(Q^\top A)$ is maximal over $Q \in \mathrm{O}(n)$ for $Q = UV^\top$. (Hint: $\operatorname{tr}(Q^\top A) = \operatorname{tr}(\Sigma Z)$ with the orthogonal matrix $Z = V^\top Q^\top U$, whose entries are at most 1 in absolute value.) This is how one aligns two point clouds by a rotation or reflection.

14. (a) Find all critical points of the monkey saddle $f(x, y) = x^3 - 3xy^2$. Show that the origin is a degenerate critical point. In polar coordinates, $f = r^3 \cos 3\varphi$; describe the regions where $f > 0$ and $f < 0$.

    (b) For which $a \in \mathbb{R}$ is $x^3 - ax$ a Morse function? Find its critical points and their indices.

    (c) Find the critical points and their indices of $f(x, y) = x^4 - 2x^2 + y^2 + \varepsilon x$ for a small $\varepsilon > 0$, and describe how the sublevel sets change as c increases.

15. Let $A = \operatorname{diag}(3, 2, 1)$ and $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$ on the sphere S^2. Find all critical points of f and their indices, and check the Morse relation $\chi(S^2) = 2$. Repeat for $A = \operatorname{diag}(4, 3, 2, 1)$ on S^3, whose Euler characteristic is 0.

16. (Alternating forms.)

    (a) For $\mathbf{u} = (1, 0, 0)$, $\mathbf{v} = (1, 2, 3)$ and $\mathbf{w} = (0, 1, 4)$, compute $(dx_1 \wedge dx_2)(\mathbf{v}, \mathbf{w})$, $(dx_1 \wedge dx_3)(\mathbf{v}, \mathbf{w})$ and $(dx_1 \wedge dx_2 \wedge dx_3)(\mathbf{u}, \mathbf{v}, \mathbf{w})$, and interpret the results as areas and a volume.

    (b) Write down a basis of $\Lambda^2(\mathbb{R}^4)^*$.

    (c) Show that $\varphi \wedge \varphi = 0$ for every 1-form φ, but that $\omega = dx_1 \wedge dx_2 + dx_3 \wedge dx_4$ satisfies $\omega \wedge \omega = 2\, dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4 \neq 0$.

17. Compute $d\omega$ for (a) $\omega = x_1x_2\, dx_1 + x_1^2\, dx_2$ on $\mathbb{R}^2$; (b) $\omega = x_2x_3\, dx_1 + x_1x_3\, dx_2 + x_1x_2\, dx_3$ on $\mathbb{R}^3$; (c) $\omega = x_1\, dx_2 \wedge dx_3 + x_2\, dx_3 \wedge dx_1 + x_3\, dx_1 \wedge dx_2$ on $\mathbb{R}^3$. Which of these forms are closed, and which are exact? Relate the result of (c) to the divergence of the vector field $F(\mathbf{x}) = \mathbf{x}$.

18. (Pullbacks.)

    (a) For the spherical coordinates $\boldsymbol{\phi}(r, \theta, \varphi) = (r\sin\theta\cos\varphi,\, r\sin\theta\sin\varphi,\, r\cos\theta)$, show that $\boldsymbol{\phi}^*(dx_1 \wedge dx_2 \wedge dx_3) = r^2 \sin\theta\, dr \wedge d\theta \wedge d\varphi$.

    (b) For the curve $\mathbf{c}(t) = (a\cos 2\pi t,\, b\sin 2\pi t)$ with $a, b > 0$, compute $\mathbf{c}^*(x_1\, dx_2 - x_2\, dx_1)$. Use Stokes' theorem for the 2-cube $(r, t) \mapsto (ar\cos 2\pi t,\, br\sin 2\pi t)$ to explain why $\frac{1}{2}\int_{\mathbf{c}}(x_1\, dx_2 - x_2\, dx_1)$ is the area πab of the ellipse. (Which faces of the 2-cube contribute to the boundary integral?)

19. Let ω be the angle form of Example A.10.25.

    (a) Show that $d\omega = 0$, and that $\eta = \frac{x_1\, dx_1 + x_2\, dx_2}{x_1^2 + x_2^2}$ is exact on $\mathbb{R}^2 \setminus \{\mathbf{0}\}$.

    (b) Compute $\int_{\mathbf{c}} \omega$ for the circle $\mathbf{c}(t) = (3 + \cos 2\pi t,\, \sin 2\pi t)$, which does not go around the origin. (Hint: on the half-plane $x_1 > 0$, $\omega = d\arctan(x_2/x_1)$.)

    (c) Show that the integral of ω over the boundary of every 2-cube in $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ is 0. Why does this not contradict $\int_{\mathbf{c}} \omega = 2\pi$ for the unit circle?

20. Verify Stokes' theorem for the 1-form $\omega = -x_2^2\, dx_1 + x_1^2\, dx_2$ and the identity 2-cube of $[0, 1]^2$ by computing both sides.

21. (Stokes' theorem on manifolds.)

(a) Let $M = \{\mathbf{x} \in \mathbb{R}^2 : 1 \le \|\mathbf{x}\| \le 2\}$ be the closed annulus, with the standard orientation. Show that M is a submanifold with boundary, describe ∂M with the induced orientation, and verify Theorem A.10.21 for $\omega = x_1\,dx_2$ by computing both sides.

(b) Let ω be a 1-form on $\mathbb{R}^3$. Show in two ways that $\int_{S^2} d\omega = 0$: with Theorem A.10.21 for S^2, and with Theorem A.10.21 for the closed upper and lower hemispheres, whose induced orientations traverse the equator in opposite directions.

22. (a) Show that $f_n(x) = \sqrt{x^2 + \frac{1}{n}}$ converges to $|x|$ uniformly on $[-1, 1]$. Conclude that the continuously differentiable functions do not form a closed subset of $C[-1, 1]$.

(b) Show that $f_n(x) = x^n$ is a Cauchy sequence in $L^2[0, 1]$, but not in $C[0, 1]$ with the maximum norm.

23. (A Neumann series.) Let V be the Volterra operator of Example A.12.5. Show that the equation $f = 1 + Vf$, i.e., $f(x) = 1 + \int_0^x f(t)\,dt$, has exactly one solution in $C[0, 1]$ (Hint: such an f solves a differential equation), and that the iteration $f_0 = 0$, $f_{k+1} = 1 + Vf_k$ gives the partial sums of the series $\sum_{j \ge 0} V^j 1$, whose terms have the norms $\frac{1}{j!}$. Which function is the solution?

24. (The best quadratic approximation.) Use the normalized Legendre polynomials of Section A.13 to find the polynomial p of degree at most 2 that minimizes $\int_{-1}^{1} (|x| - p(x))^2\,dx$, and compute the minimal value.

25. Compute the Fourier series of $f(x) = x^2$ on $[-\pi, \pi]$, and use Parseval's identity to show that $\sum_{k=1}^{\infty} \frac{1}{k^4} = \frac{\pi^4}{90}$. Why do the partial sums converge uniformly here, unlike for $f(x) = x$? (Hint: compare the decay of the coefficients.)

26. (Riesz representers.)

(a) Let A be symmetric positive definite, and give $\mathbb{R}^n$ the inner product $\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^\top A \mathbf{y}$. Find the vector that represents the functional $\mathbf{x} \mapsto \mathbf{c}^\top \mathbf{x}$.

(b) Find the function in $L^2[0, 1]$ that represents $f \mapsto \int_0^{1/2} f(x)\,dx$.

(c) Show that $f \mapsto f(\frac{1}{2})$ is not bounded on $C[0, 1]$ with the norm of $L^2[0, 1]$. (Hint: narrow tents.)

27. Check by integrating that $e_1(t) = \sqrt{2} \sin \frac{\pi t}{2}$ is an eigenfunction of the integral operator of Example A.14.4 with the eigenvalue $\frac{4}{\pi^2}$, and that $\|e_1\|_{L^2} = 1$.

28. (Minimal-norm interpolation.) Let $0 < t_1 < \cdots < t_N \le 1$ and $y_1, \ldots, y_N \in \mathbb{R}$. In the RKHS of the kernel $\min(s, t)$ on $[0, 1]$ (Example A.15.2), find the function f with the smallest norm such that $f(t_i) = y_i$ for all i. (Hint: as in the proof of the representer theorem, $f = \sum_i \alpha_i k_{t_i}$.) Show that f is the piecewise linear function through the points $(0, 0), (t_1, y_1), \ldots, (t_N, y_N)$, constant on $[t_N, 1]$.

29. For the linear kernel $k(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{y}$, show that the prediction $\mathbf{k}_*^\top (K + \lambda I)^{-1} \mathbf{y}$ of kernel ridge regression is $\mathbf{x}_*^\top \hat{\mathbf{w}}$ with the ridge regression solution $\hat{\mathbf{w}} = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}$, where X has the rows $\mathbf{x}_i^\top$. (Compare with the exercise on Gaussian processes and Bayesian linear regression in Lecture 4.)

Programming Exercises

As in Lecture 3, use NumPy (and Matplotlib for plots), and start each file with

```
import numpy as np
import matplotlib.pyplot as plt
rng = np.random.default_rng(0)
```

P1. Inverting a residual layer. Let `W = rng.normal(size=(5, 5))` and $\mathbf{g}(\mathbf{x}) = c\tanh(W\mathbf{x})$ with $c = 0.9/\|W\|_2$.

- Show that $\mathbf{g}$ is Lipschitz continuous with constant 0.9 (use $|\tanh'| \le 1$ and Lemma A.5.1).
- For a random $\mathbf{x}$, compute $\mathbf{y} = \mathbf{x}+\mathbf{g}(\mathbf{x})$, and recover $\mathbf{x}$ with the iteration of Corollary A.5.5. Plot the error against the number of iterations on a logarithmic scale, together with the bound of Theorem A.5.3.
- Repeat with $c = 1.5/\|W\|_2$. Does the iteration still converge? Is $\mathbf{x} \mapsto \mathbf{x} + \mathbf{g}(\mathbf{x})$ still injective? (Try to find two points with the same image.)

P2. Riemannian gradient ascent on the sphere. For a random symmetric $A = B+B^\top$ with `B = rng.normal(size=(6, 6))`, maximize $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$ on S^5 with the Riemannian gradient and the normalization retraction (Section A.8).

- Start from 100 random points, and check that you always reach $\pm\mathbf{q}_1$ (compare with `np.linalg.eigh`). Plot $f(\mathbf{p}_k)$ for a few runs.
- Start exactly at an eigenvector $\mathbf{q}_i$ with $i > 1$. What happens? Now add a perturbation of size 10^{-8}. How many steps does it take to escape from the saddle point, and how does this depend on $\lambda_1 - \lambda_i$?

P3. Orthogonal Procrustes. Generate a point cloud `X = rng.normal(size=(50, 3))`, a random orthogonal matrix Q_0 (from `np.linalg.qr`), and $Y = XQ_0^\top + 0.01\cdot\text{noise}$. Recover Q_0 by maximizing $\operatorname{tr}(Q^\top Y^\top X)$ over $\mathrm{O}(3)$ with the SVD formula of Exercise 13. Then solve the same problem with Riemannian gradient ascent on $\mathrm{O}(3)$: project $G = \nabla_Q$ onto $T_Q\mathrm{O}(3)$ by $Q\,\frac{1}{2}(Q^\top G - G^\top Q)$, take a step, and retract with the Q factor of a QR decomposition (fixing the signs so that the diagonal of R is positive). Compare the results.

P4. Counting components of sublevel sets. For $f(x, y) = x^4 - 2x^2 + y^2 + 0.3x$, evaluate f on a fine grid over $[-2, 2]^2$, and for 50 values of c, count the connected components of the grid points with $f \le c$ (write a breadth-first search over neighboring grid points). Plot the number of components against c, and compare the values of c where it changes with the critical values of f, which you can find with Newton's method applied to $\nabla f = \mathbf{0}$.

P5. Integrating differential forms.

- Write a function that computes $\int_{\mathbf{c}} \omega$ for a 1-form $\omega = \sum_i F_i\,dx_i$ on $\mathbb{R}^n$ and a curve $\mathbf{c} : [0, 1] \to \mathbb{R}^n$, using the pullback $\int_0^1 F(\mathbf{c}(t))^\top \mathbf{c}'(t)\,dt$ and the midpoint rule with N points. Pass F, $\mathbf{c}$ and $\mathbf{c}'$ as Python functions.
- For the angle form of Example A.10.25, compute $\frac{1}{2\pi}\int_{\mathbf{c}} \omega$ for the unit circle, for the circle of radius 1 around $(3, 0)$, for the unit circle traversed twice, $\mathbf{c}(t) = (\cos 4\pi t, \sin 4\pi t)$, and for the curve $\mathbf{c}(t) = \big((2 + \cos 10\pi t)\cos 2\pi t,\ (2 + \cos 10\pi t)\sin 2\pi t\big)$. Check that the results are close to integers, and explain them.

- Verify Stokes' theorem numerically for the 2-cube $\mathbf{c}(u,v) = (u, v, u^2 - v^2)$ in $\mathbb{R}^3$ and the 1-form $\omega = x_3\,dx_1 + x_1\,dx_2$: compute $\int_{\mathbf{c}} d\omega$ from the pullback of $d\omega = dx_3 \wedge dx_1 + dx_1 \wedge dx_2$, and $\int_{\partial \mathbf{c}} \omega$ as the signed sum of four line integrals. Both should be 2.

P6. Picard iteration. Consider the logistic equation $x' = x(1 - x)$ with $x(0) = 0.2$, whose solution is $x(t) = \frac{1}{1+4e^{-t}}$.

- On a grid of 1001 points in $[0, 5]$, implement the Picard iteration $x_{k+1}(t) = 0.2 + \int_0^t x_k(s)(1 - x_k(s))\,ds$ of Theorem A.12.4, starting from $x_0(t) = 0.2$ and computing the integrals with the cumulative trapezoidal rule. Plot the iterates $x_0, \ldots, x_8$ together with the solution.
- Plot the maximum error of x_k against k on a logarithmic scale, on $[0, 1]$ and on $[0, 5]$. Where does the iteration converge first, and why?

P7. Fourier series and the Gibbs phenomenon.

- Plot the partial sums of the Fourier series of $f(x) = x$ on $[-\pi, \pi]$ with 5, 20 and 100 terms (Example A.13.7), and compute their errors in the L^2 norm and in the maximum norm. Which error tends to 0?
- For $N = 50, 100, 200, 400, 800$ terms, compute the maximum of the partial sum on a fine grid near π, and the overshoot above π as a fraction of the jump 2π. Compare with the limit ≈ 0.0895.
- Check Parseval's identity numerically, and compare the partial sums of $\sum_k \frac{1}{k^2}$ with $\frac{\pi^2}{6}$. How fast do they converge?

P8. The Karhunen–Loève expansion. Let $t_i = \frac{i}{n}$ for $i = 1, \ldots, n$ with $n = 500$.

- The matrix $\frac{1}{n}\big(\min(t_i, t_j)\big)_{i,j}$ approximates the integral operator of Example A.14.4 (why?). Compute its largest eigenvalues with `np.linalg.eigh`, and compare them with $\frac{1}{(k-1/2)^2\pi^2}$. Compare the eigenvectors, scaled by $\sqrt{n}$, with $\sqrt{2}\sin\big((k - \frac{1}{2})\pi t\big)$.
- Sample Brownian paths from the expansion truncated after 5, 20 and 200 terms, and compare them with paths simulated as random walks (Lecture 4). Check $\mathbb{E}[B(0.5)^2] = 0.5$ for the truncated expansions with 10^4 samples.

P9. Kernel ridge regression.

- Implement kernel ridge regression with the squared exponential kernel (Example A.15.4), and check that it gives the posterior mean of your Gaussian process regression in the programming exercises of Lecture 4, with $\lambda = \sigma^2$. Plot the fits for $\lambda = 10^{-4}$, 10^{-2} and 1.
- For the polynomial kernel $k(x, y) = (1 + xy)^2$ on $\mathbb{R}$, show that $k(x, y) = \boldsymbol{\phi}(x)^\top \boldsymbol{\phi}(y)$ with $\boldsymbol{\phi}(x) = (1, \sqrt{2}x, x^2)$, and check numerically that kernel ridge regression gives the same predictions as ridge regression with these features. Compare the running times of the two methods for $N = 10^2, 10^3, 10^4$ data points.